

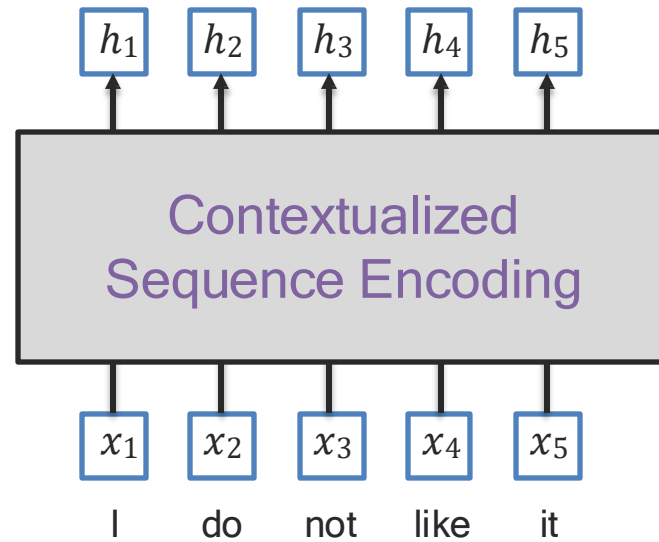


Lecture 6:

Multimodal Transformers

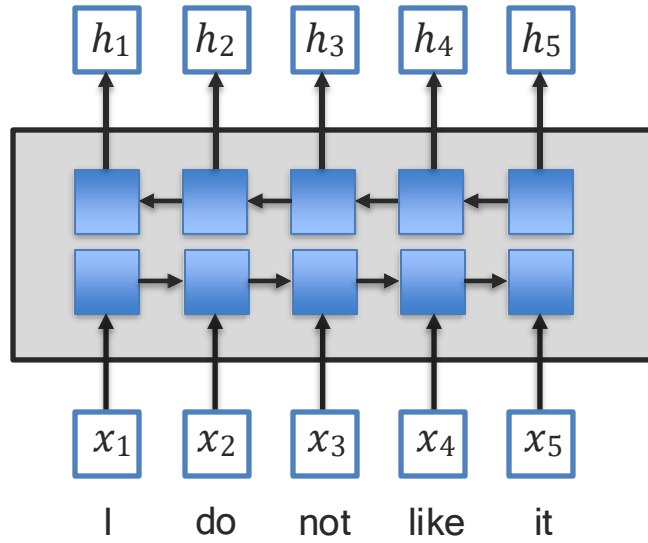
Email: idanschwartz@gmail.com or
idan.schwartz@biu.ac.il

Sequence Encoding



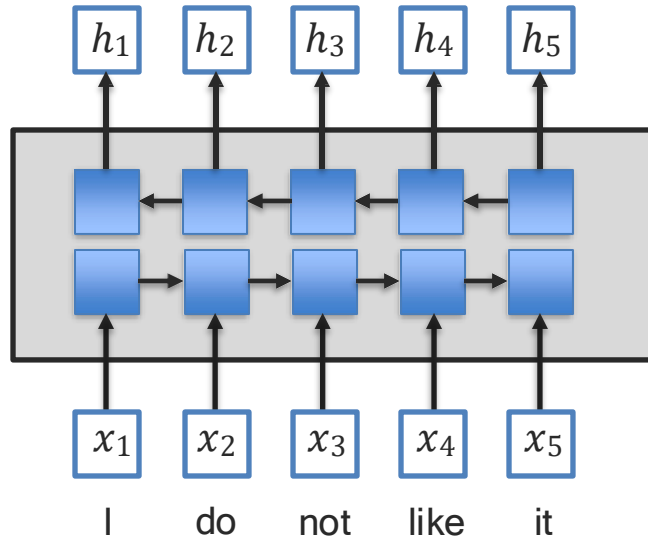
How to encode this sequence while modeling the interaction between elements (e.g., words)?

Option 1: Bi-directional LSTM



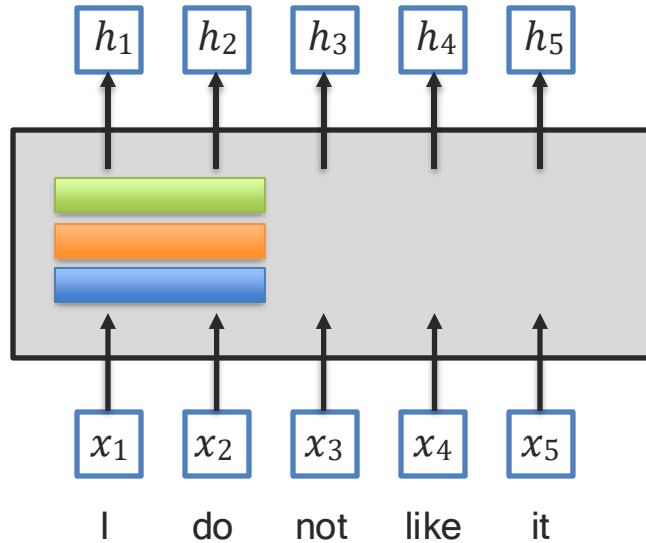
But harder to parallelize...

Option 1: Bi-directional LSTM



But harder to parallelize...

Option 2: Convolutions

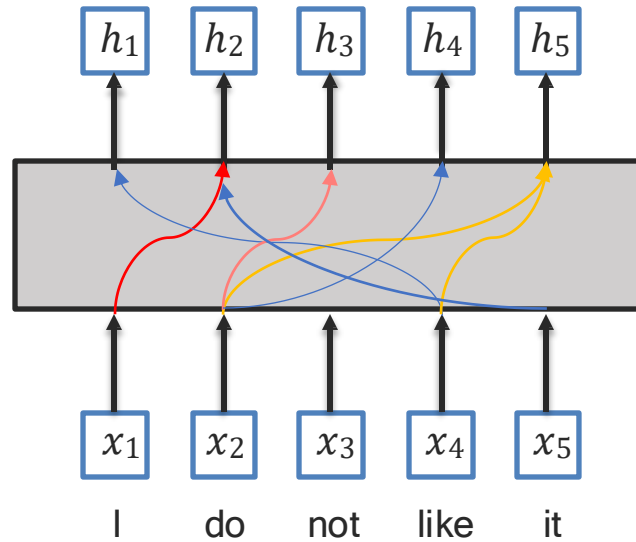


Can be parallelized!

But modeling long-range dependencies
require multiple layers

And convolutional kernels are static

Option 3: Self-attention



Can be parallelized!

Long-range dependencies

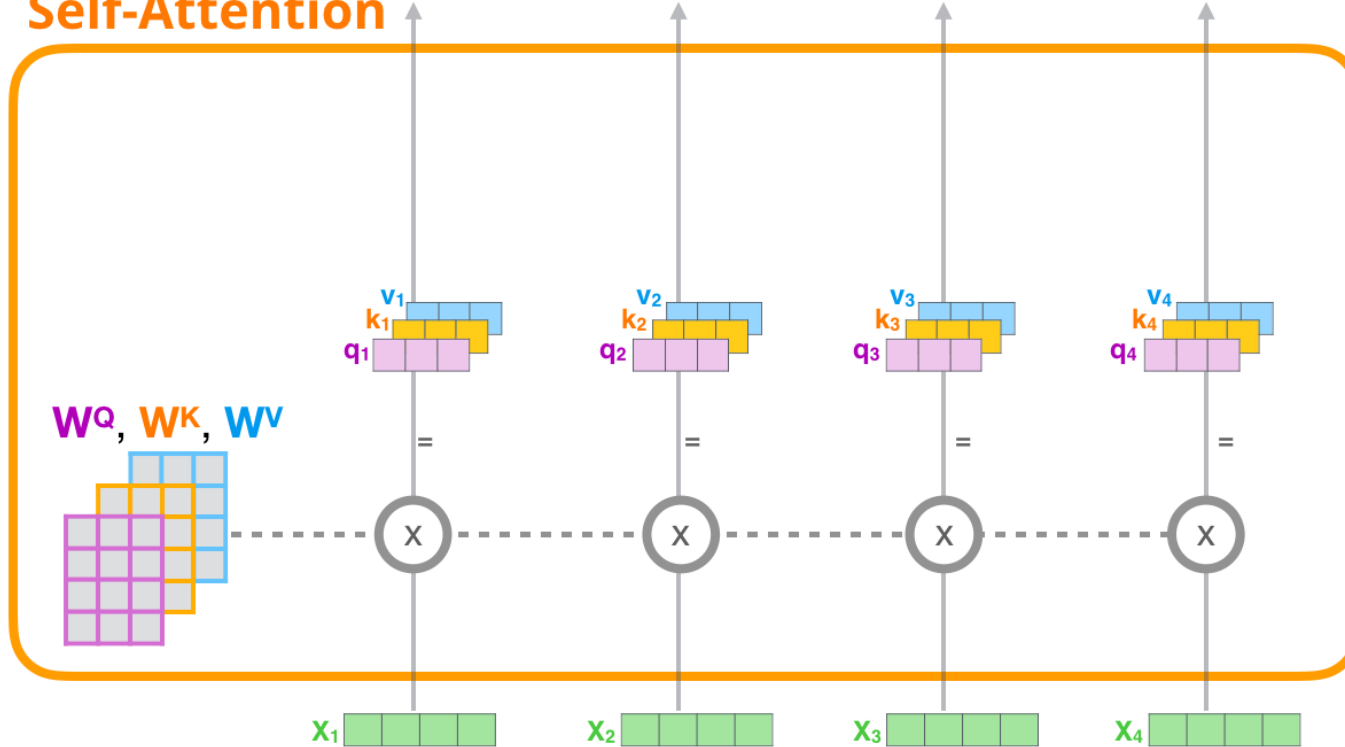
Dynamic attention weights

All you need...

Transformer

1) For each input token, create a **query vector**, a **key vector**, and a **value vector** by multiplying by weight Matrices W^Q , W^K , W^V

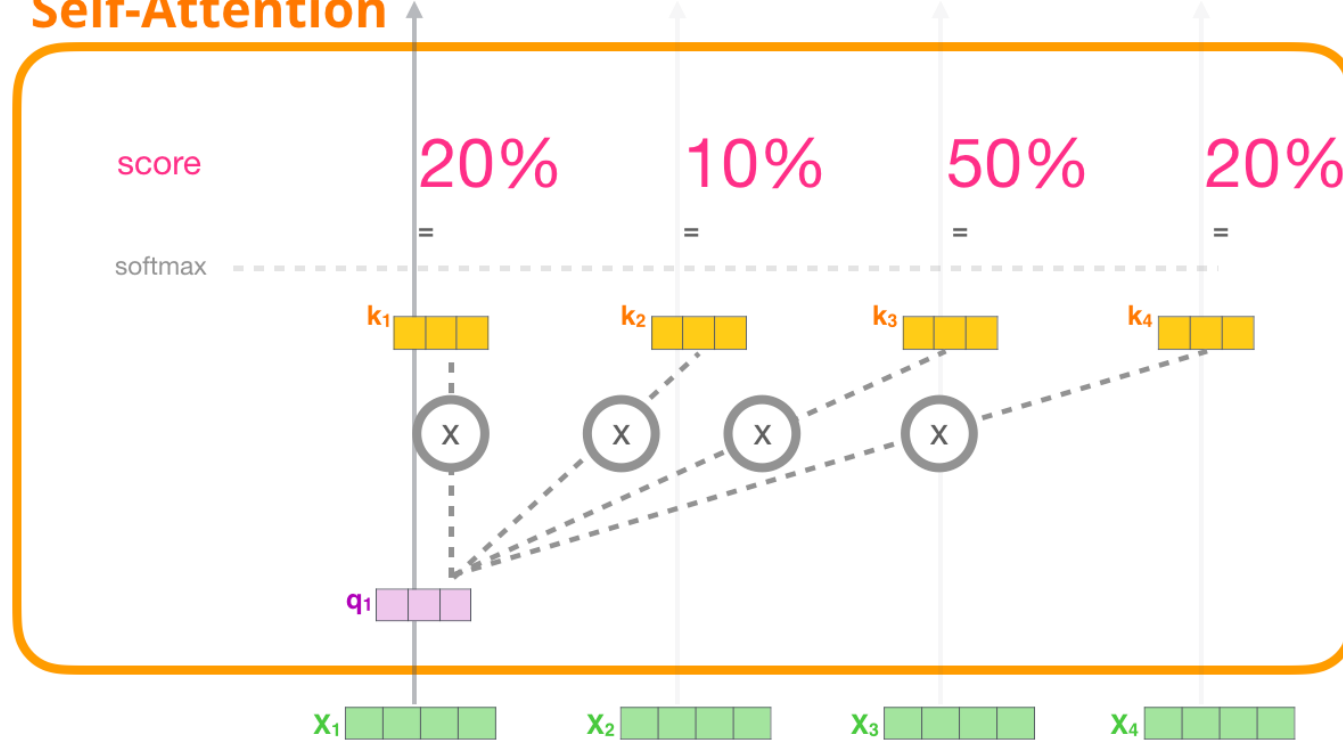
Self-Attention



Transformer

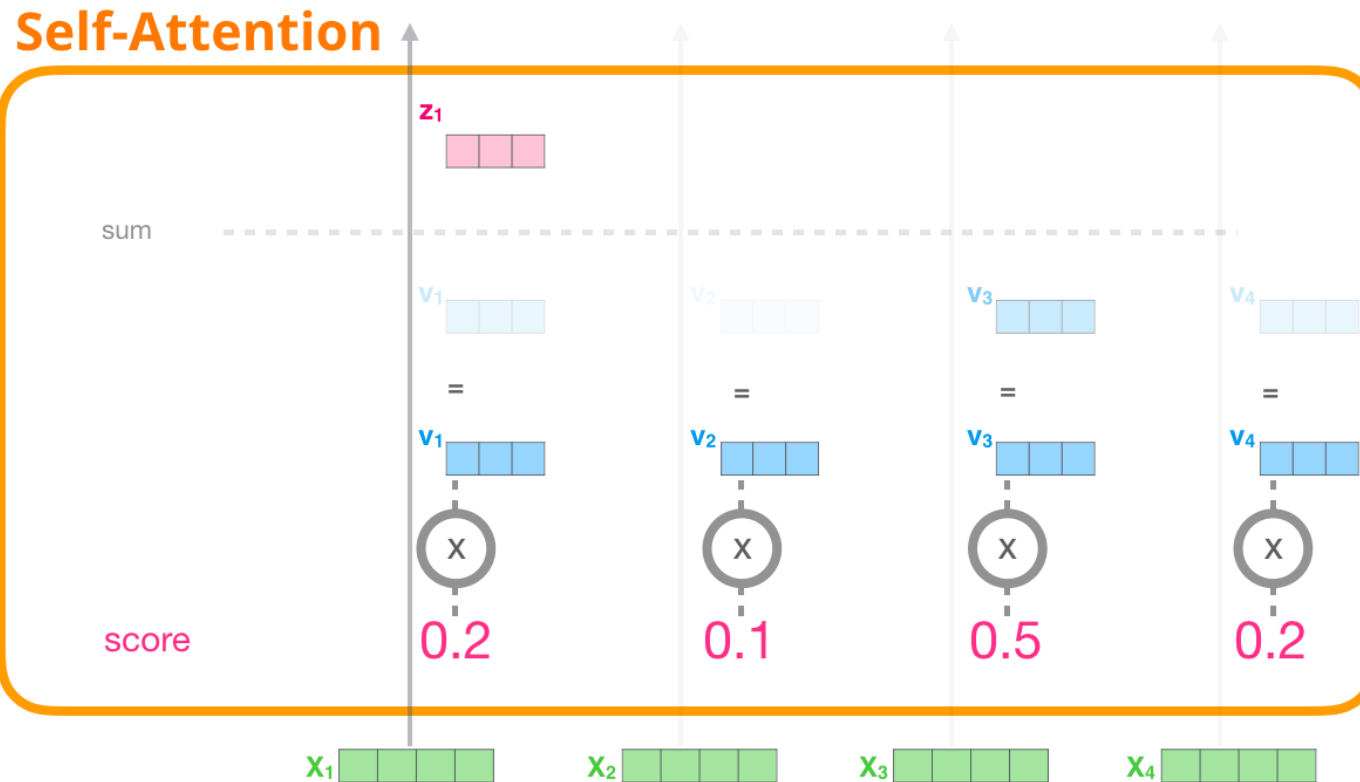
2) Multiply (dot product) the current **query vector**, by all the **key vectors**, to get a score of how well they match

Self-Attention



Transformer

3) Multiply the **value vectors** by the **scores**, then sum up



Transformer

- Given a representation $x \in \mathbb{R}^{n \times d}$
- The attention score is based on a **dot-product**
 - $Q = x \cdot W_Q \in \mathbb{R}^{n \times d}$
 - $K = x \cdot W_K \in \mathbb{R}^{n \times d}$
 - Where: $W_Q \in \mathbb{R}^{d \times d}$, $W_K \in \mathbb{R}^{d \times d}$

Correlation Matrix: $\mathbf{Q} \times \mathbf{K}^T \in \mathbb{R}^{n \times n}$

Self-correlation matrix

Correlation Matrix: $\overset{\text{Q}}{\begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline \end{array}} \times \overset{\text{K}^T}{\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline & \\ \hline \end{array}} \in \mathbb{R}^{n \times n}$

	What	animal	is
What	0.4	0.1	0.1
animal	0.1	0.3	0.1
is	0.1	0.3	0.3

Transformer

- Given a sentence $x \in \mathbb{R}^{n \times d}$
 - Dimensions:
 - $Q = x \cdot W_Q \in \mathbb{R}^{n \times d}$
 - $K = x \cdot W_K \in \mathbb{R}^{n \times d}$
 - $V = x \cdot W_V \in \mathbb{R}^{n \times d}$
 - $W_Q, W_K, W_v \in \mathbb{R}^{d \times d}$

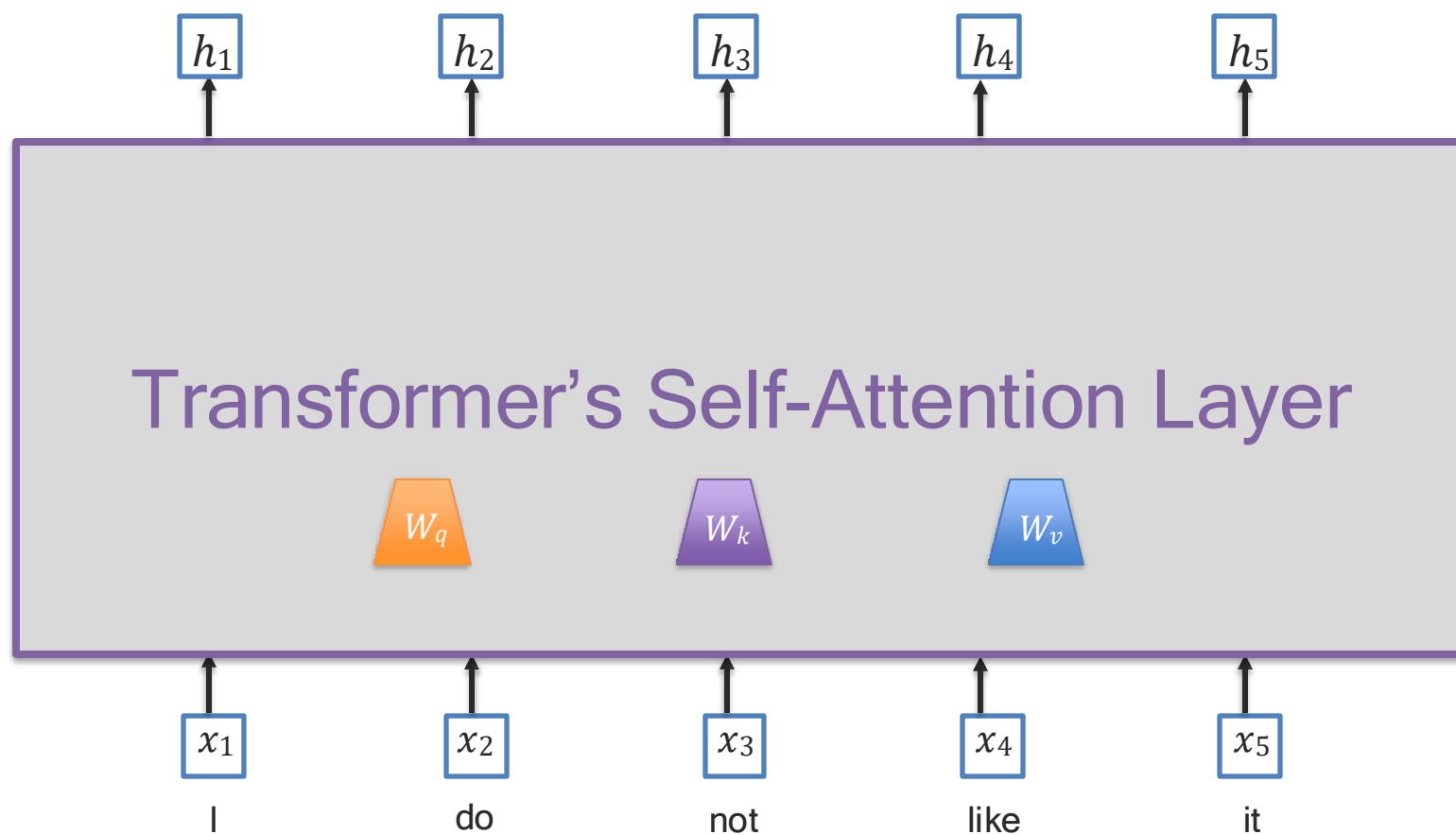
Self-correlation matrix

$$\text{softmax} \left(\frac{Q \times K^T}{\sqrt{d_k}} \right) V$$

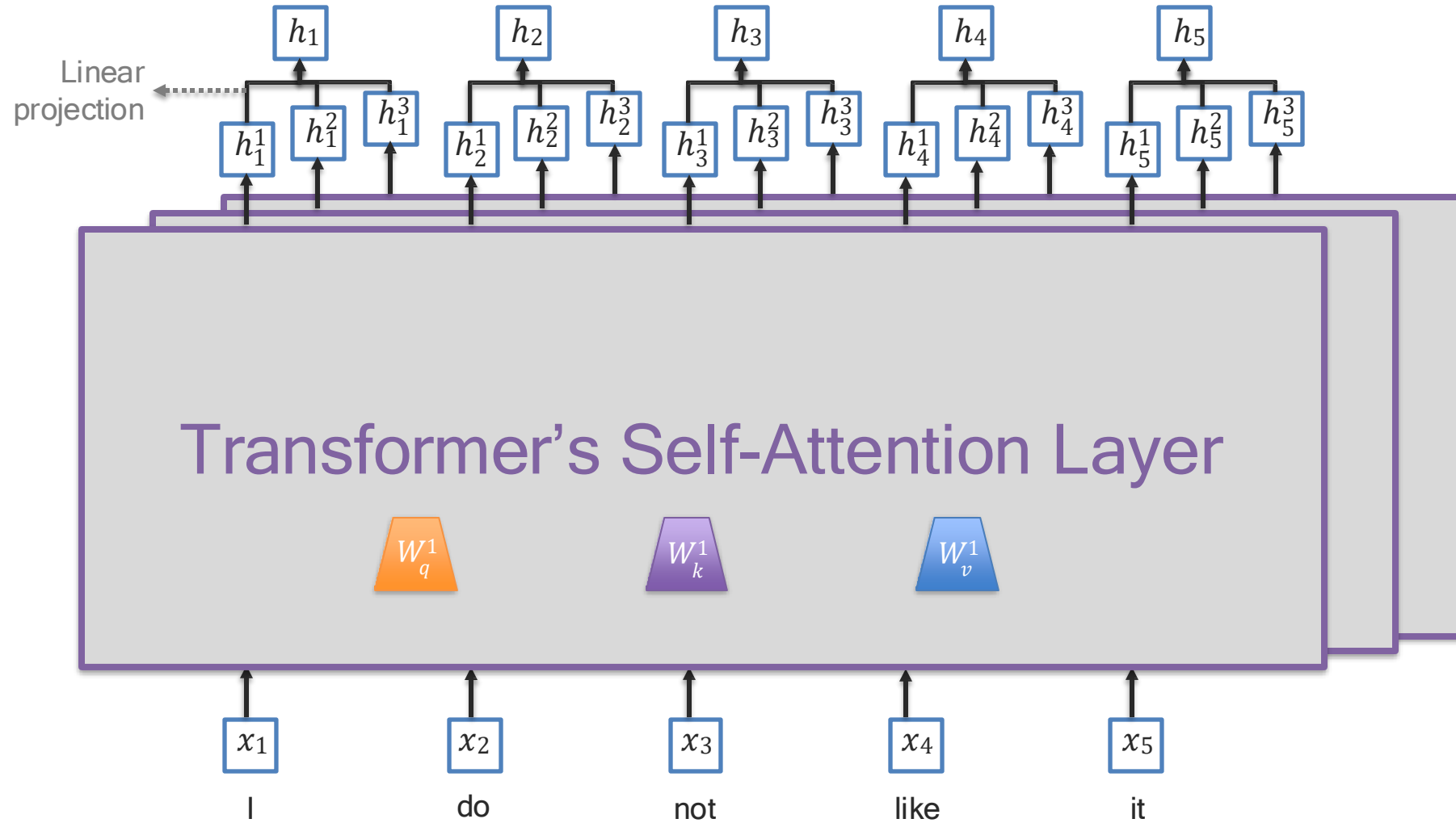
Z

The diagram illustrates the self-attention mechanism in a Transformer. It shows the calculation of the self-attention matrix Z as the softmax of the scaled dot product of the query matrix Q and the key matrix K^T , multiplied by the value matrix V . The matrices are represented as grids: Q is a 2x3 purple grid, K^T is a 3x2 orange grid, V is a 2x3 blue grid, and Z is a 2x3 pink grid. A bracket labeled 'Self-correlation matrix' spans the Q and K^T matrices.

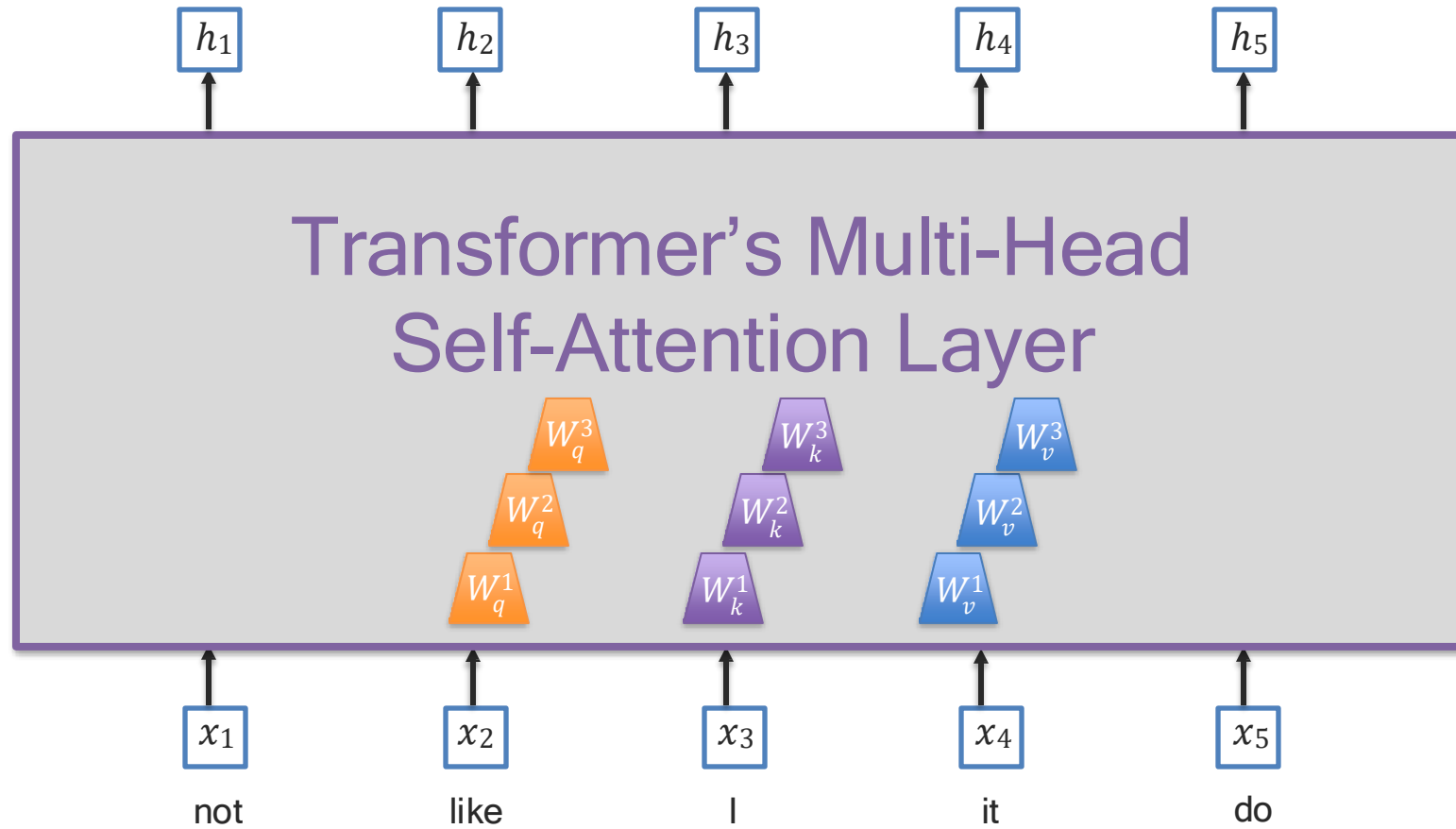
Transformer



Transformer Multi-Head Self-Attention

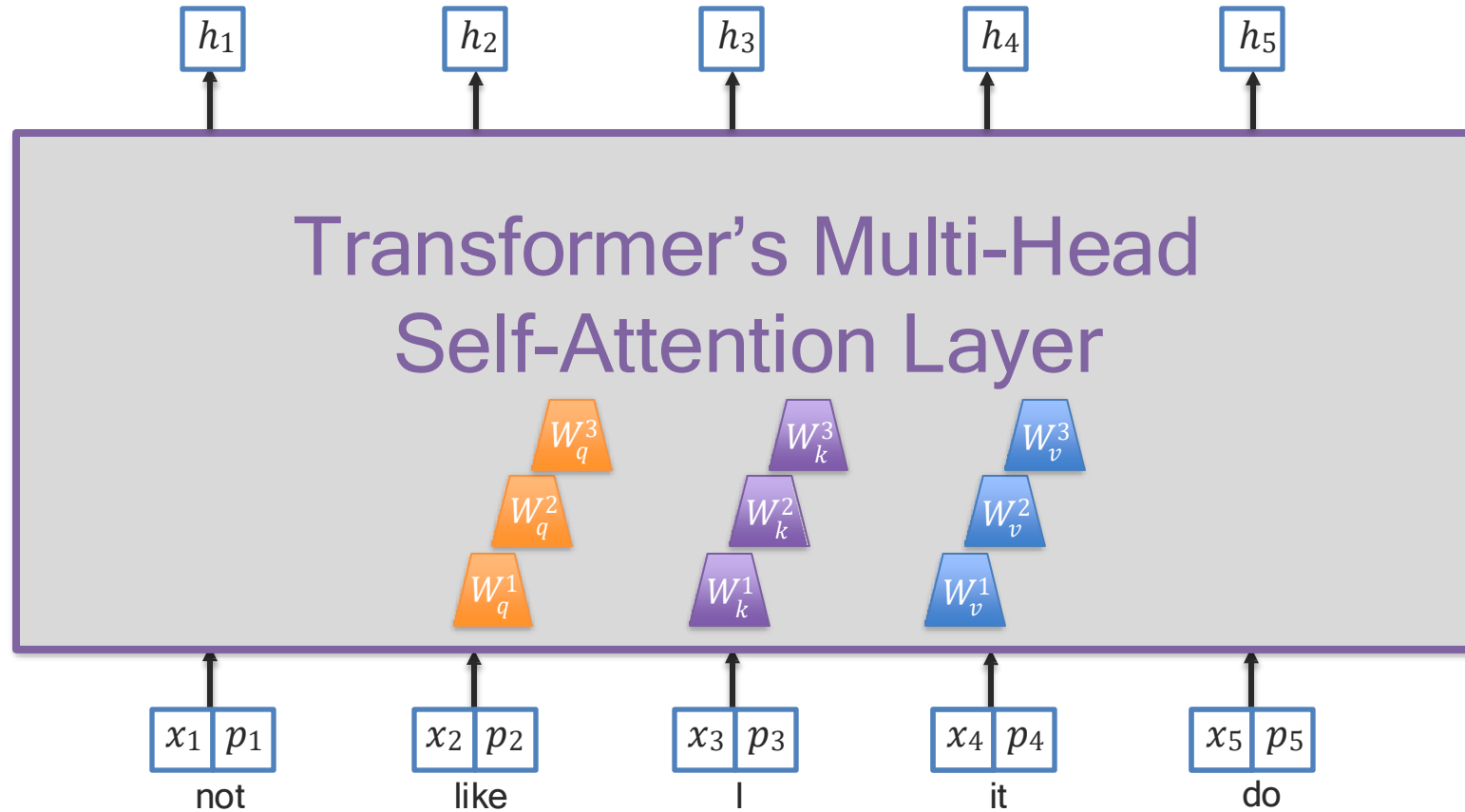


Transformer Multi-Head Self-Attention



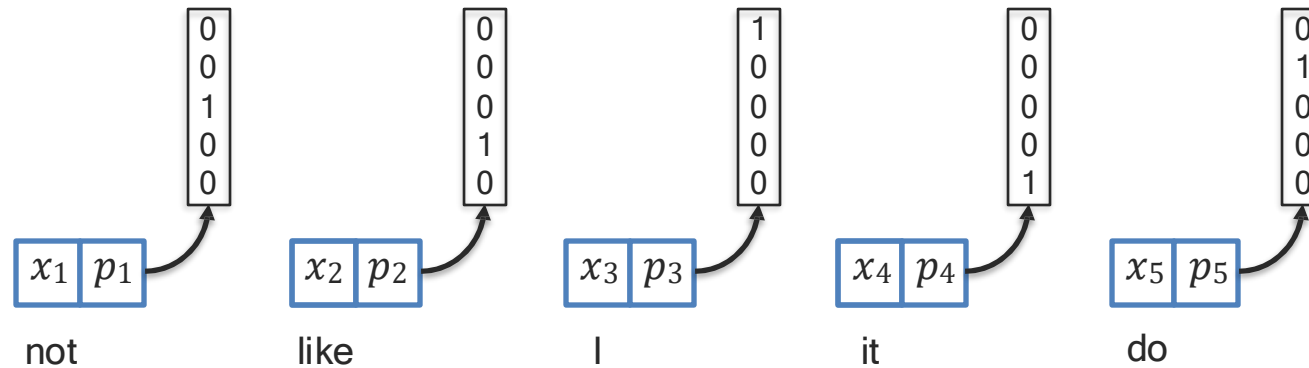
What happens if the words are shuffled?

Transformer + Positional Encoding

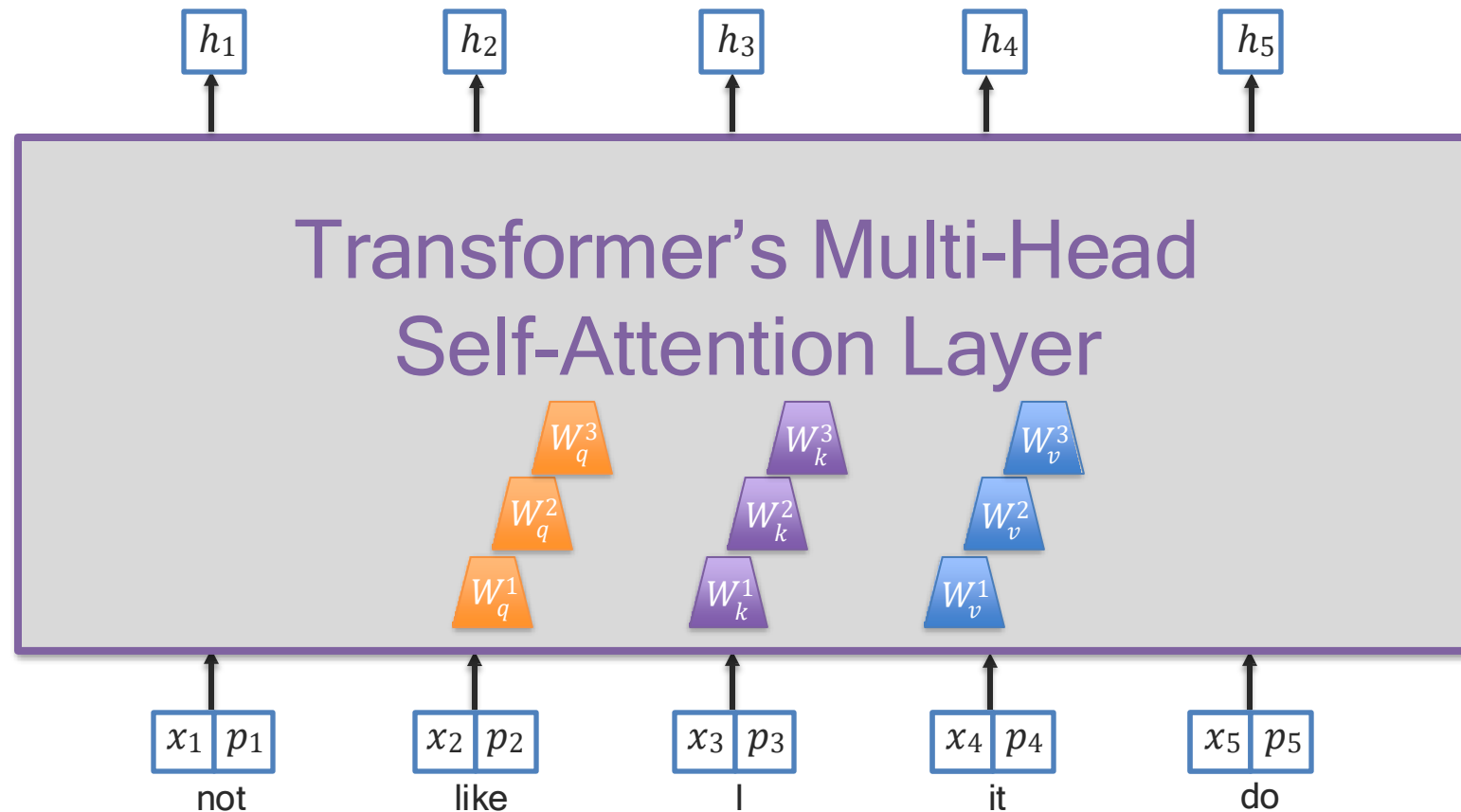


Position embeddings

- Position information is not encoded in a self-attention module
- **E.g.**, one-hot encoding

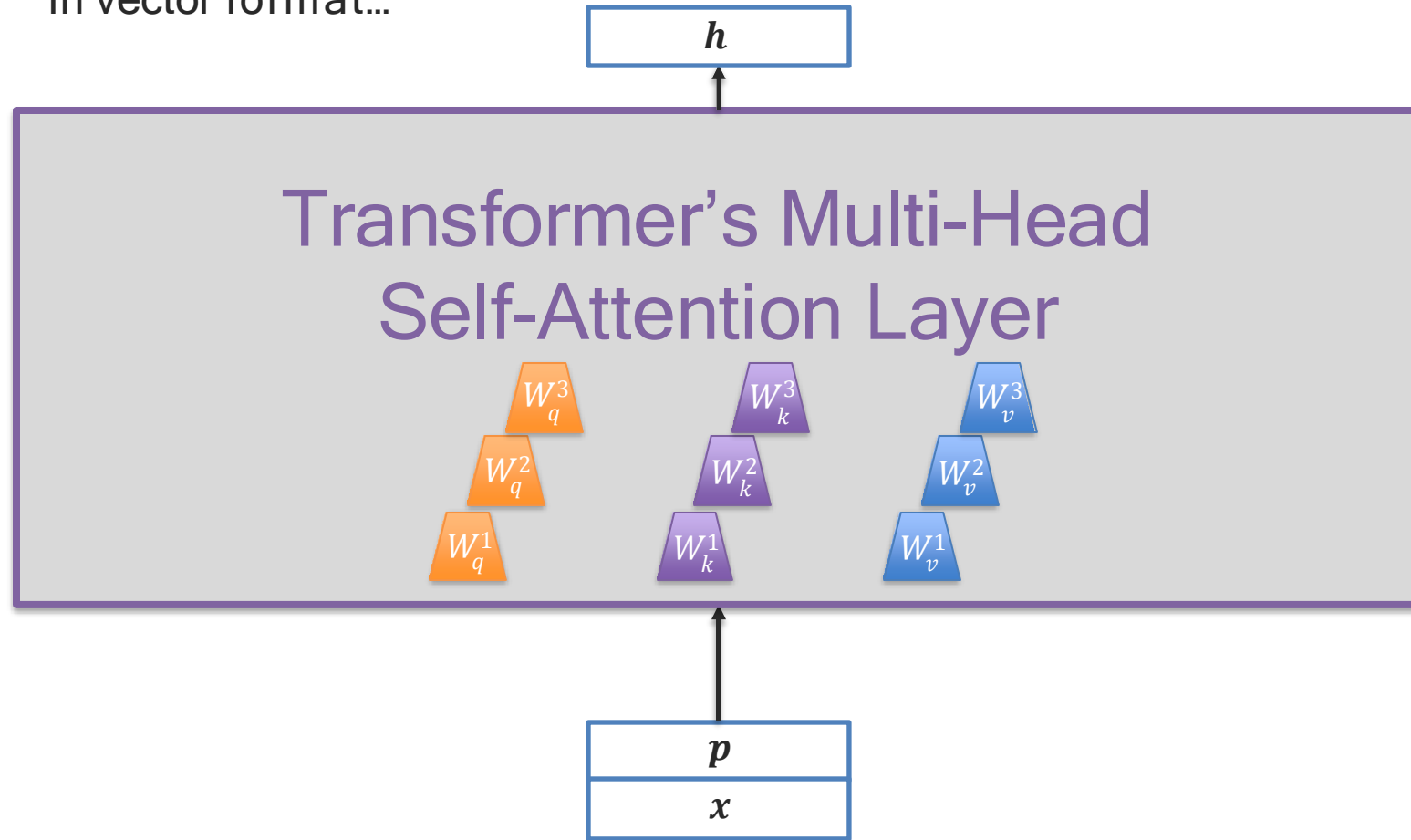


Transformer Multi-Head Self-Attention

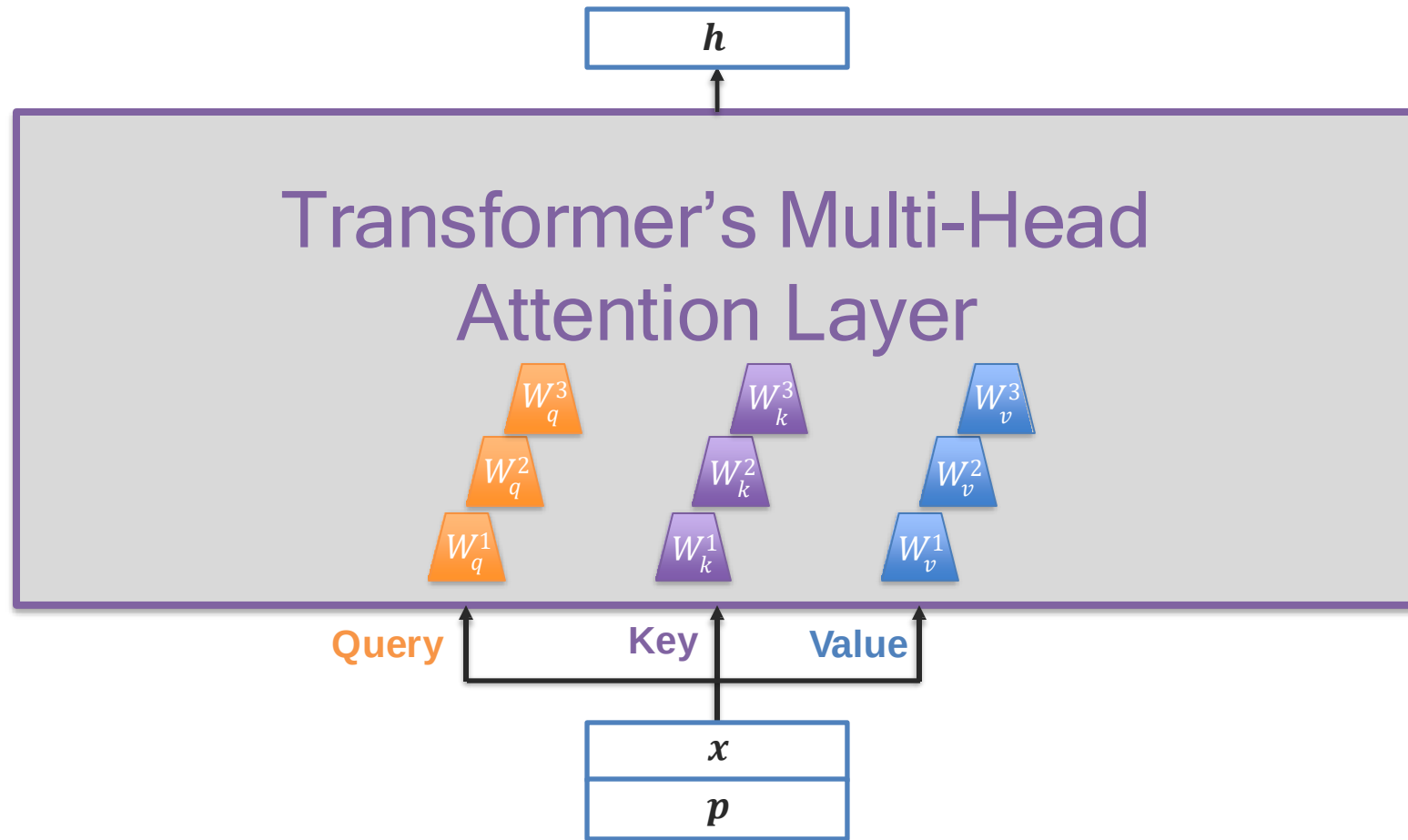


Transformer Multi-Head Self-Attention

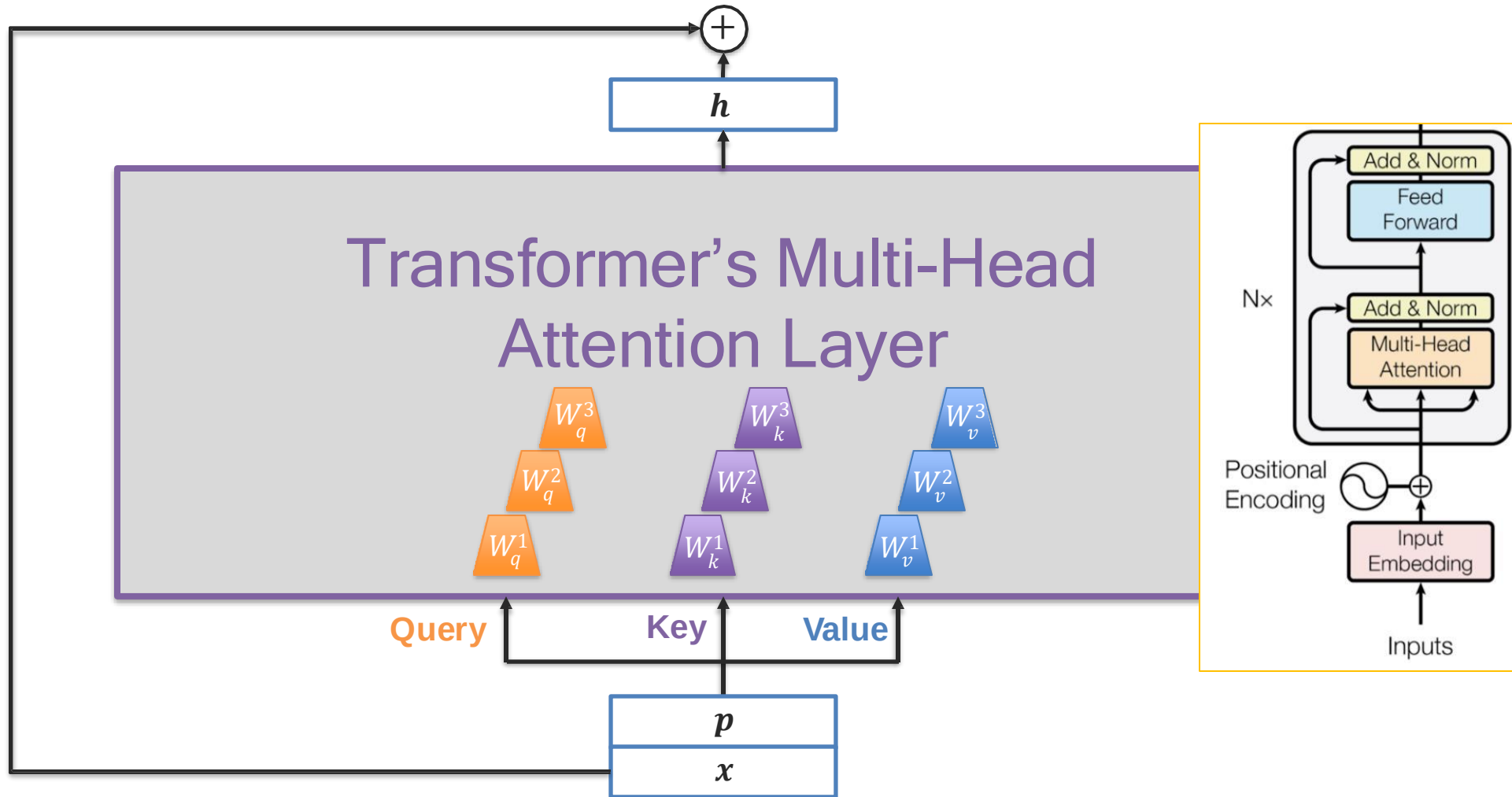
In vector format...



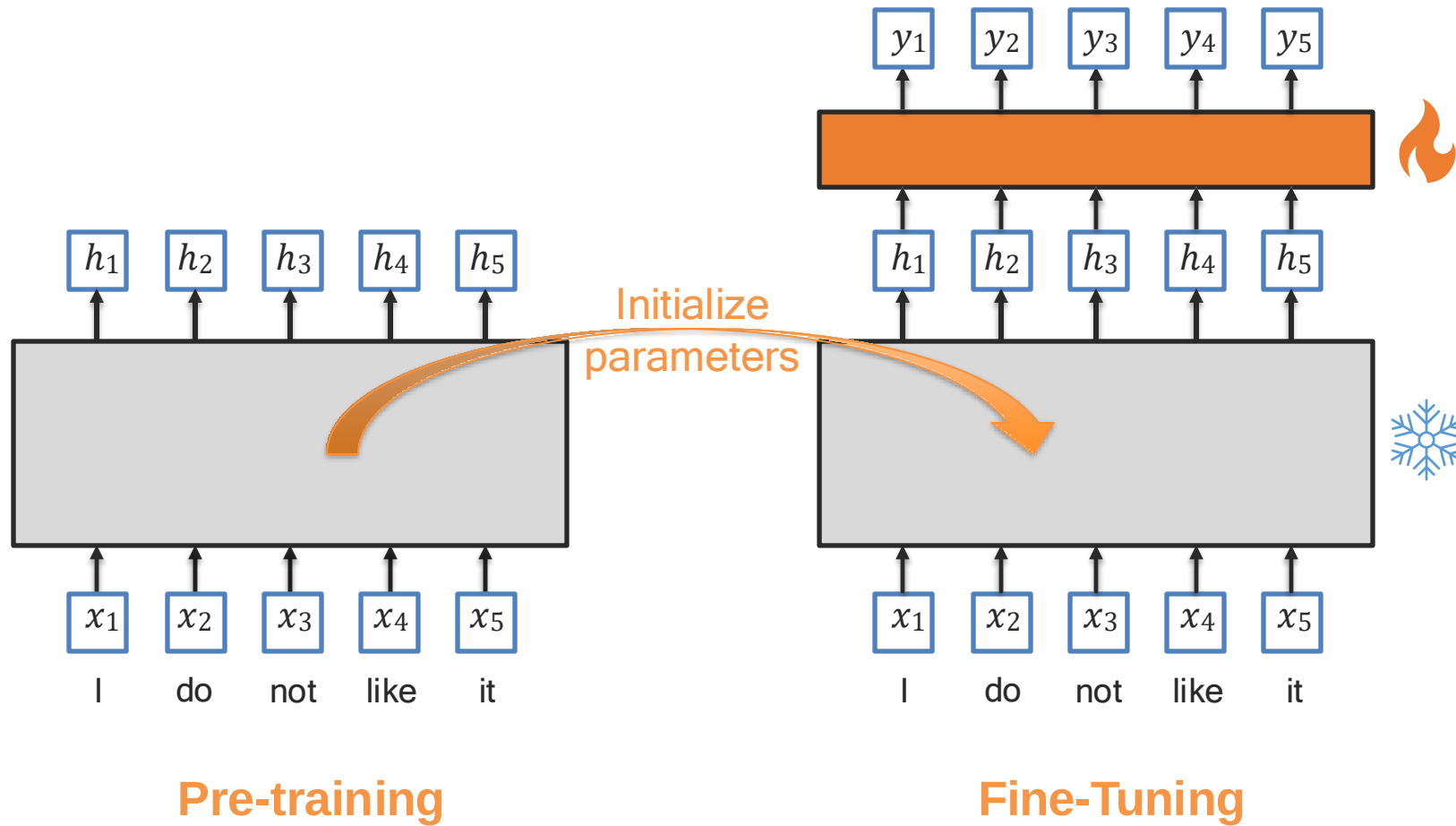
Transformer Multi-Head Self-Attention



Transformer – Residual Connection



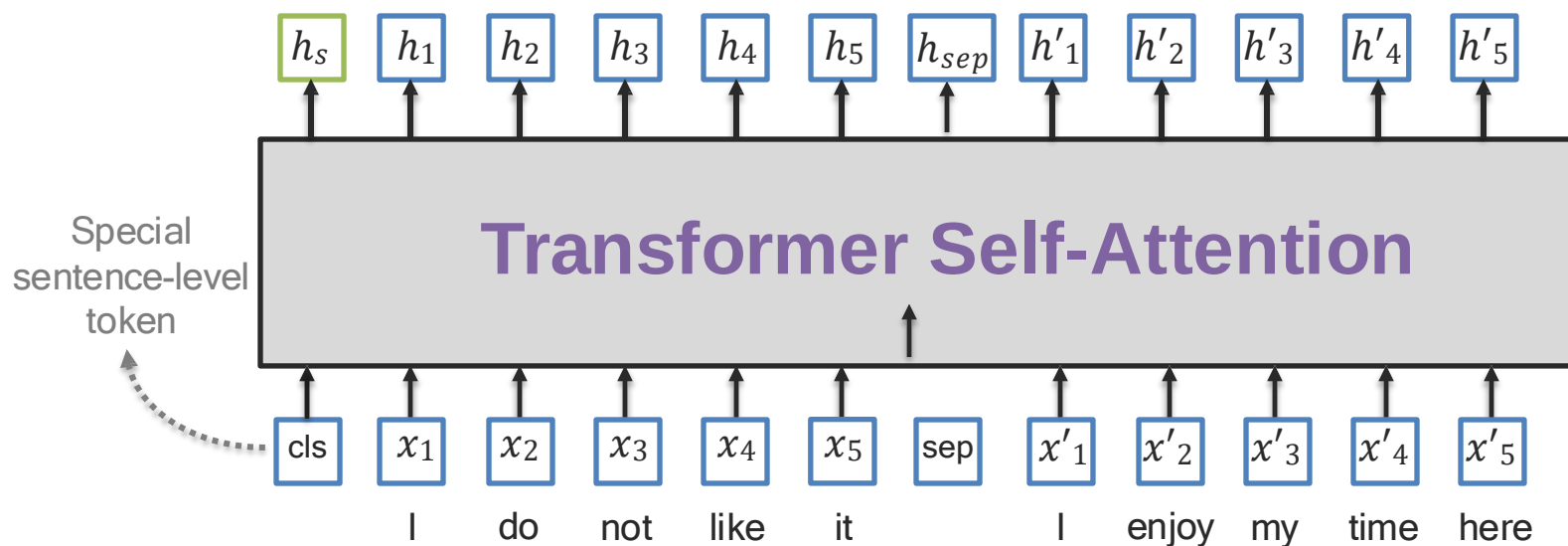
Pre-Training and Fine-Tuning



BERT: Bidirectional Encoder Representations

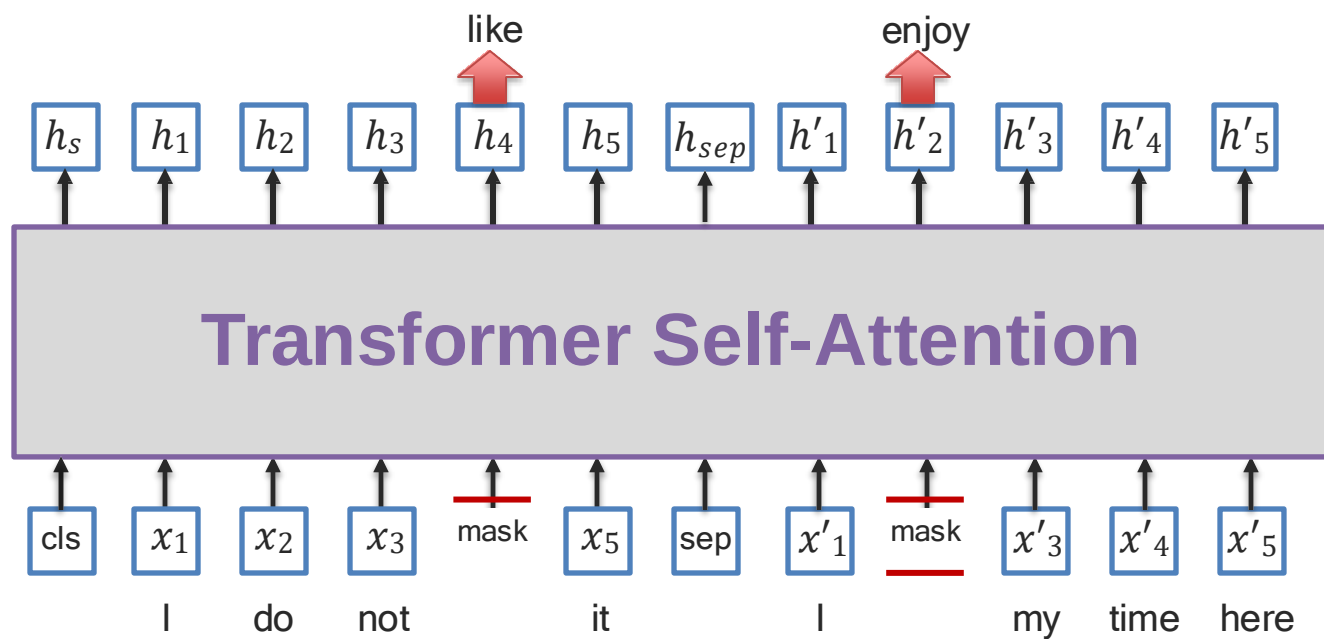
Advantages:

- 1 Jointly learn representation for token-level and sentence level
- 2 Same network architecture for pre-training and fine-tuning
- 3 Can be used learn relationship between sentences
- 4 Models bidirectional and long-range interactions between tokens



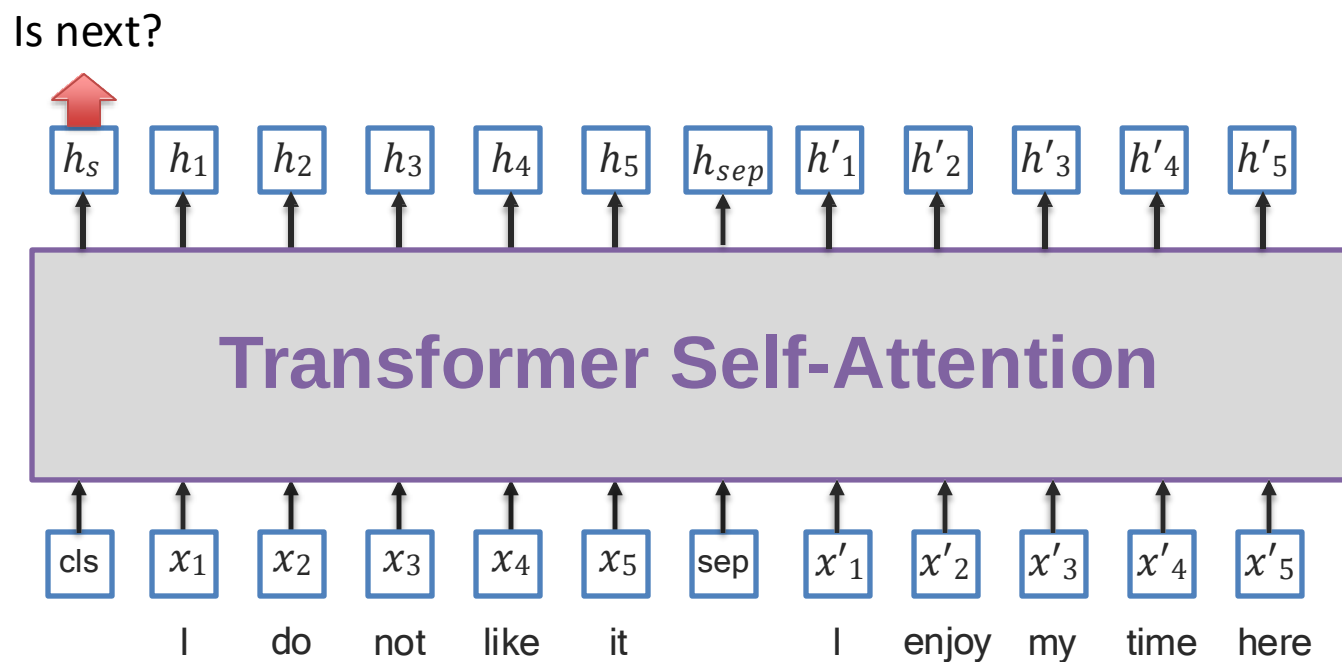
Self-supervised Pre-training

- **Masked Language Model:** Randomly mask input tokens and then try to predict them



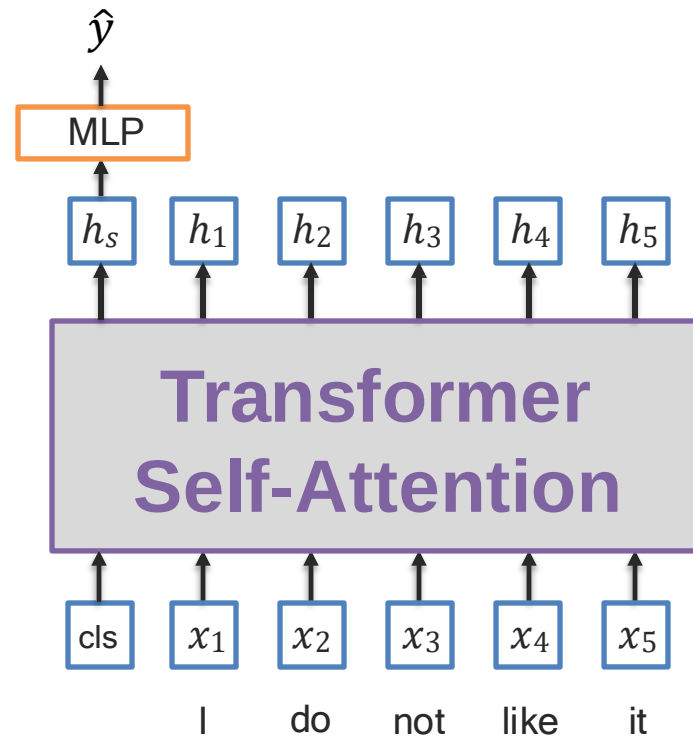
Self-supervised Pre-training

- **Next Sentence Prediction:** Given two sentences, predict if this is the next one or not



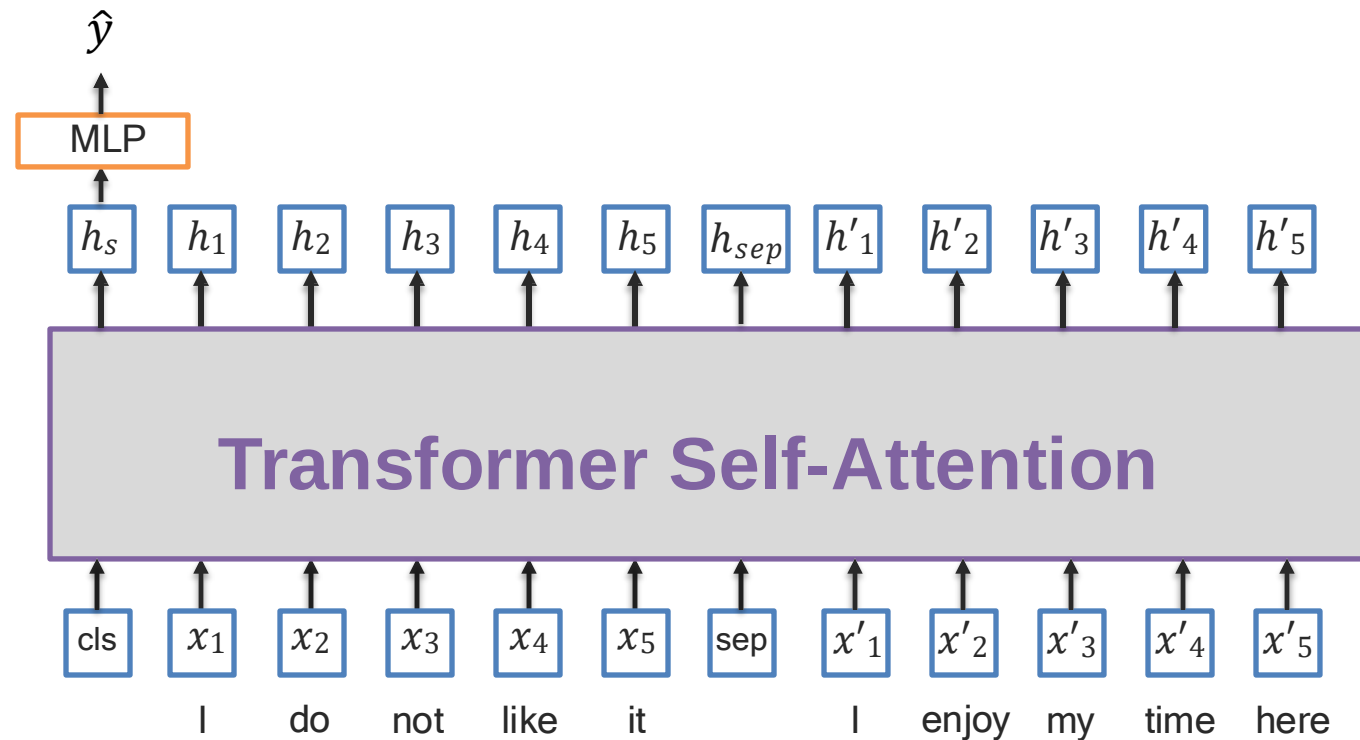
Fine-tuning

- Sentence-level classification



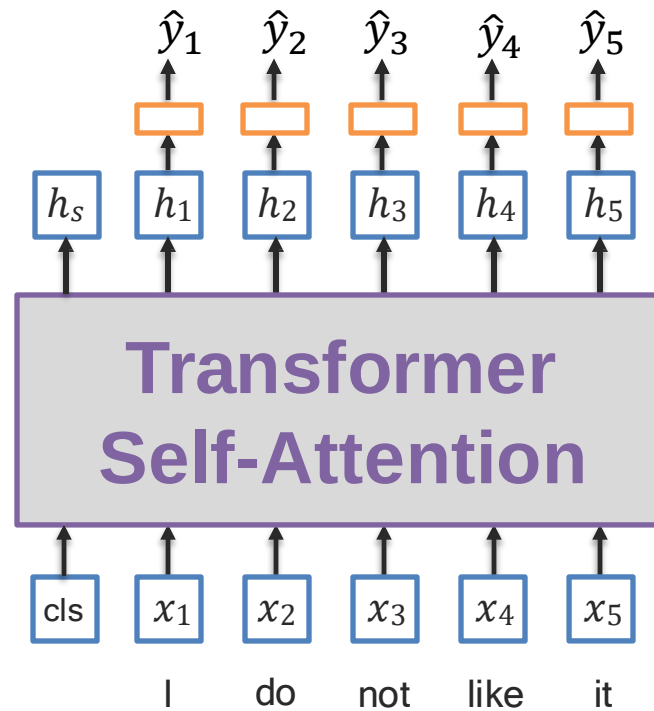
Fine-tuning

- More than one sentence



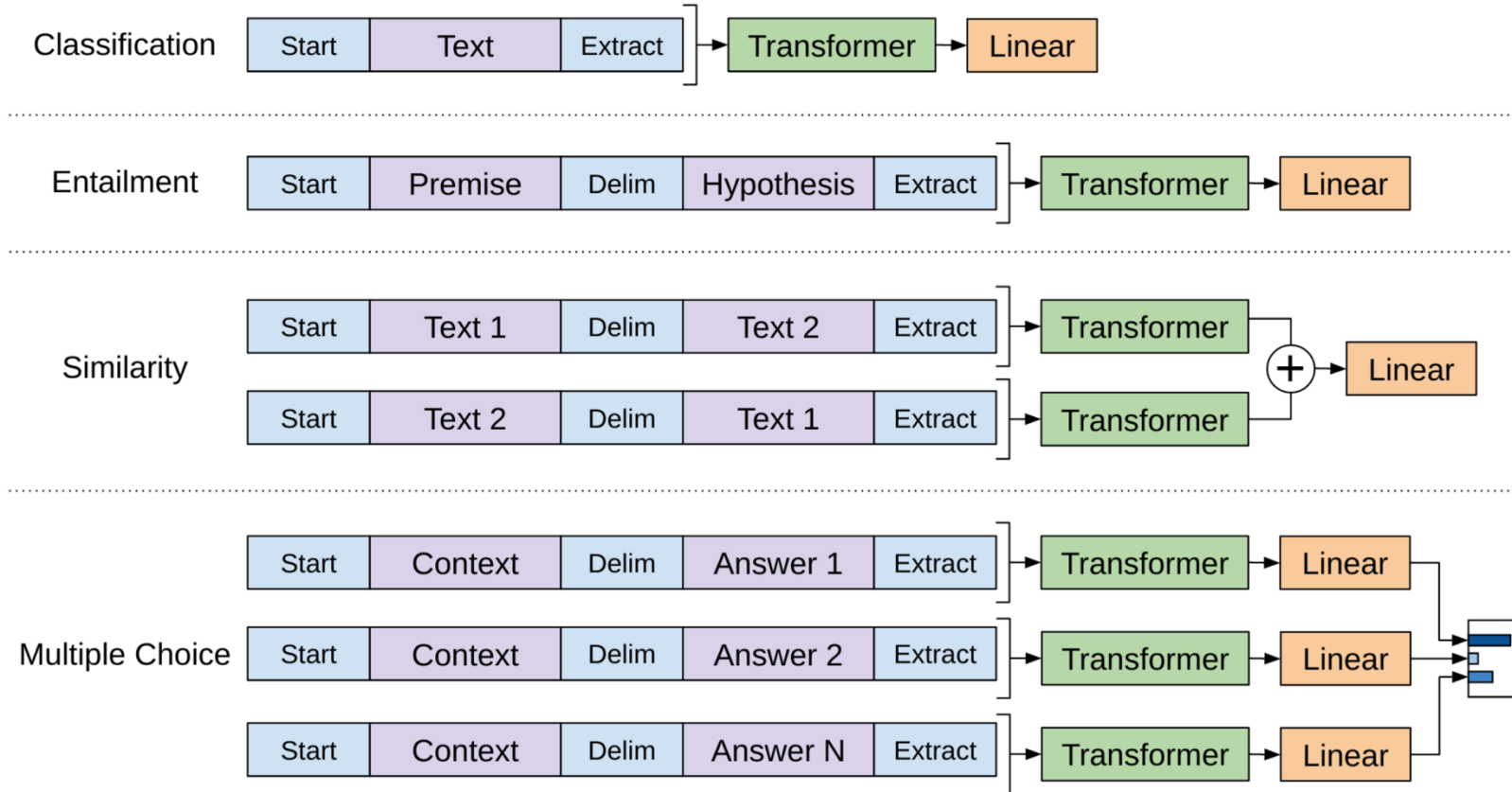
Fine-tuning

- Token-level classification

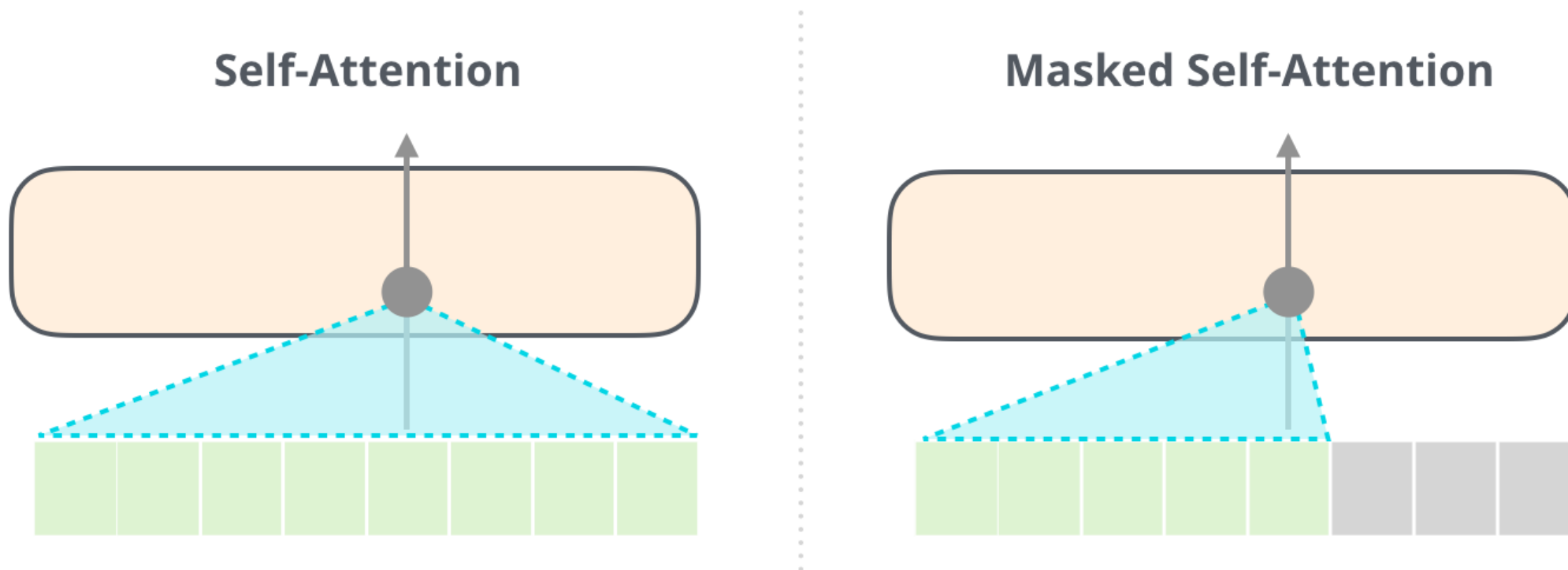


Auto-regressive is all you need

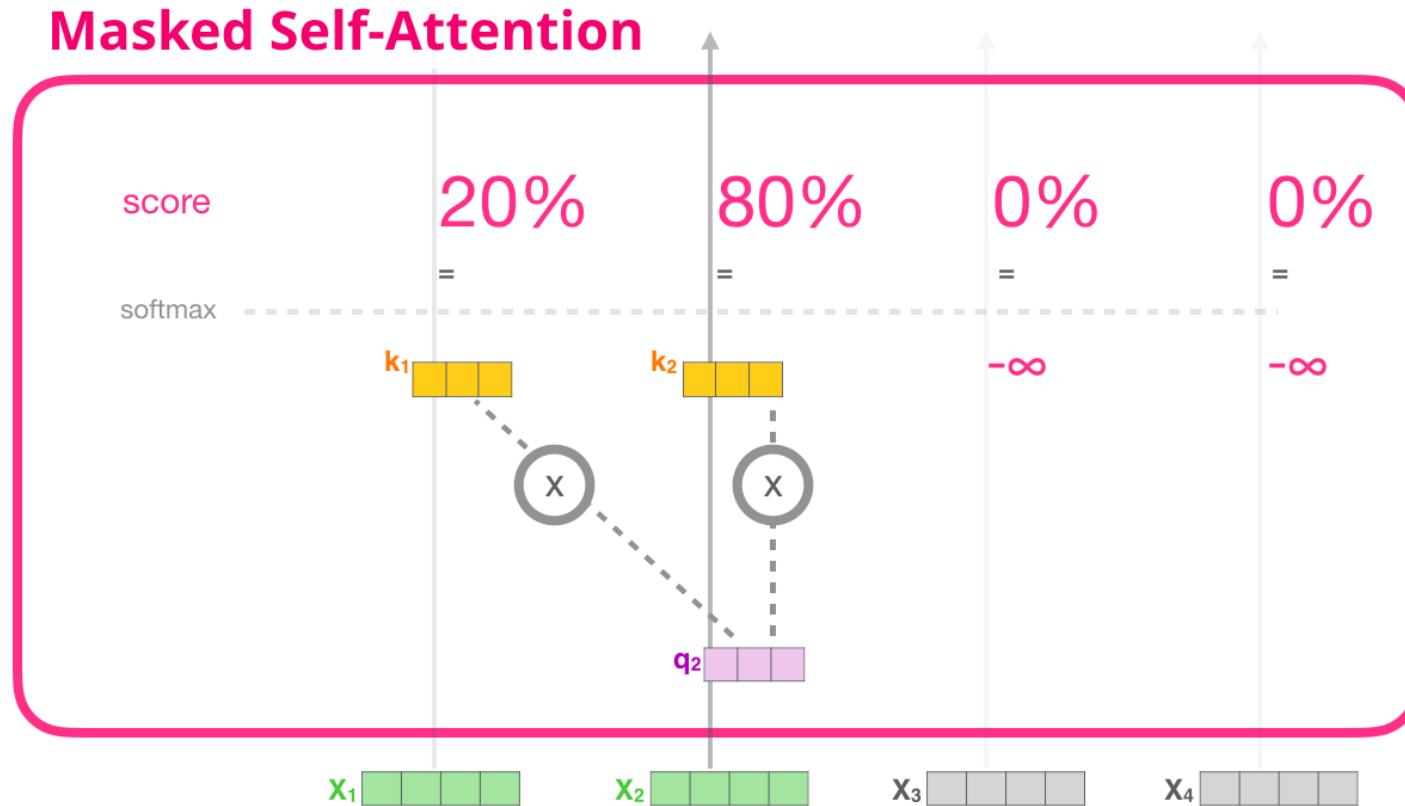
Other Fine-tuning Approaches



Transformer decoding: masked attention

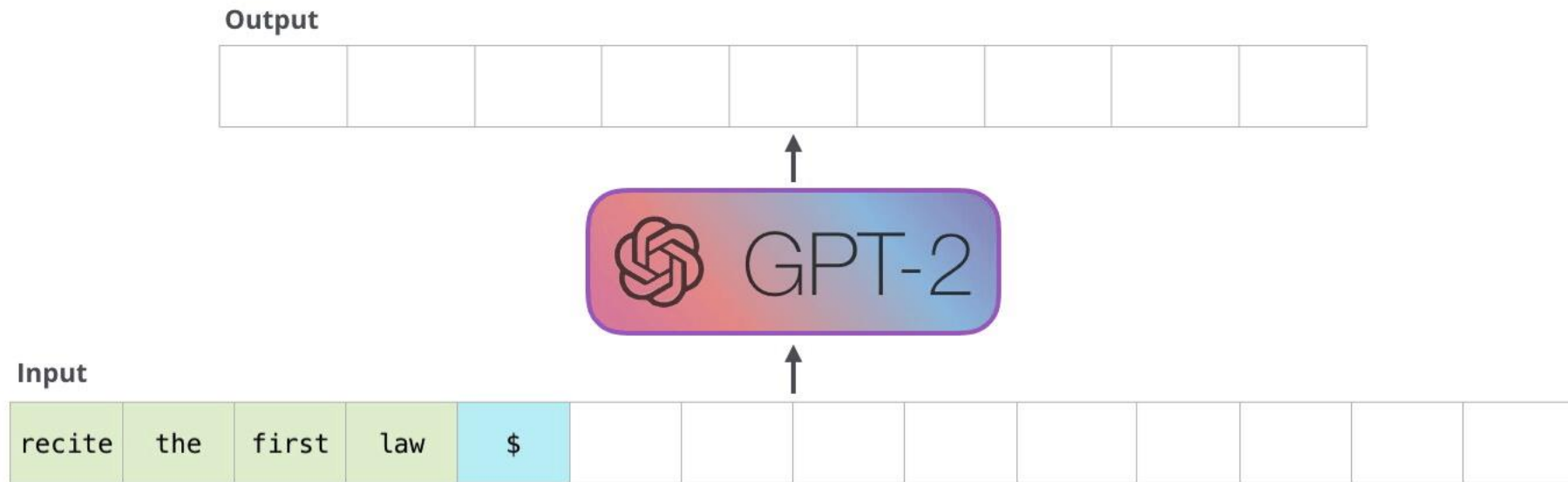


Transformer decoding: masked attention



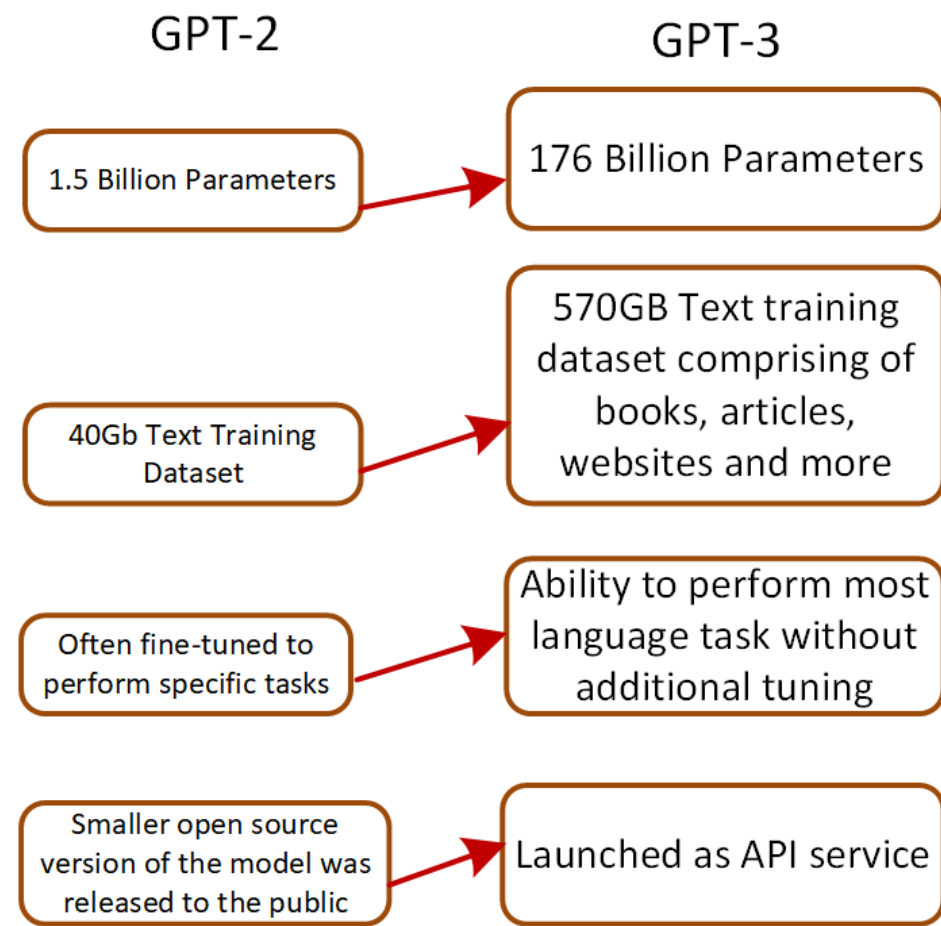
Transformer encoder-decoder

GPT-2 showed - all you need is an autoregressive decoding



GPT-3; Easy to scale

Model Name	n_{params}	n_{layers}	d_{model}	n_{heads}	d_{head}
GPT-3 Small	125M	12	768	12	64
GPT-3 Medium	350M	24	1024	16	64
GPT-3 Large	760M	24	1536	16	96
GPT-3 XL	1.3B	24	2048	24	128
GPT-3 2.7B	2.7B	32	2560	32	80
GPT-3 6.7B	6.7B	32	4096	32	128
GPT-3 13B	13.0B	40	5140	40	128
GPT-3 175B or “GPT-3”	175.0B	96	12288	96	128

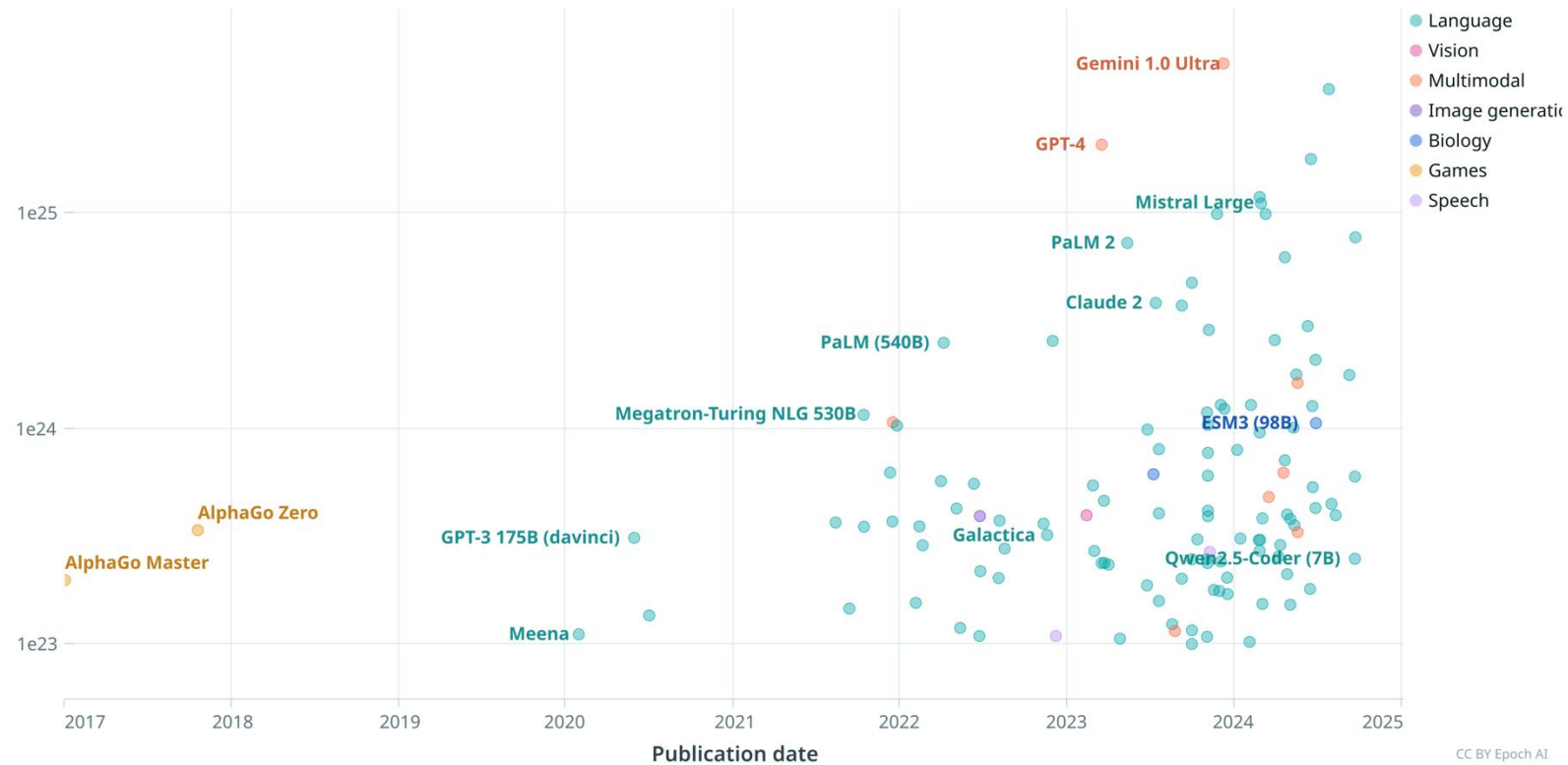


LLMs Moore's Law

Large-Scale AI Models

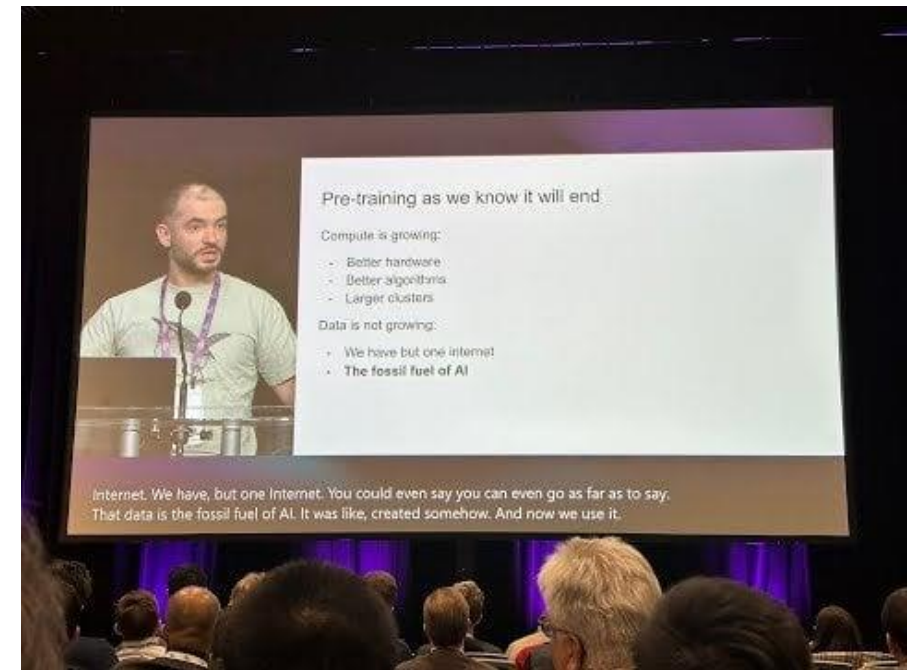
EPOCH AI

Training compute (FLOP)



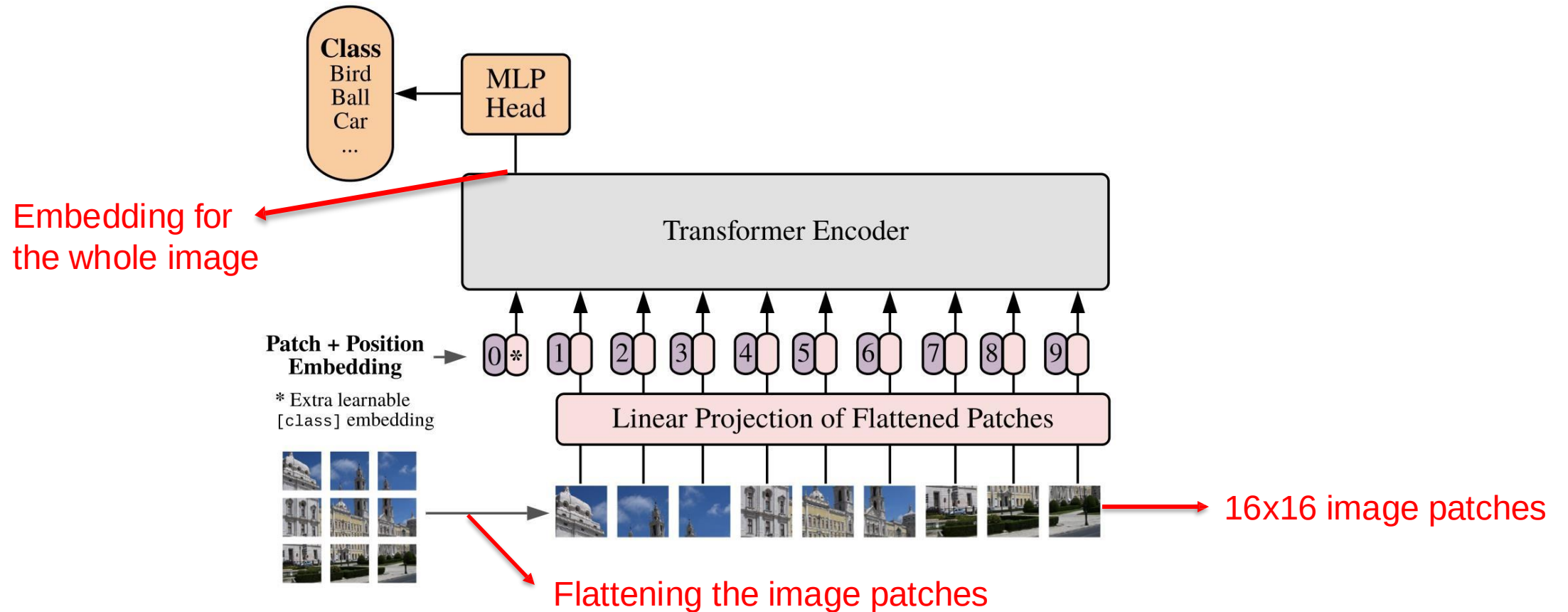
Can pre-training continue?

- Compute is growing:
 - Better hardware
 - Better algorithms
 - Larger clusters
- Data is not growing:
 - We have but one internet
 - **The fossil fuel of AI**



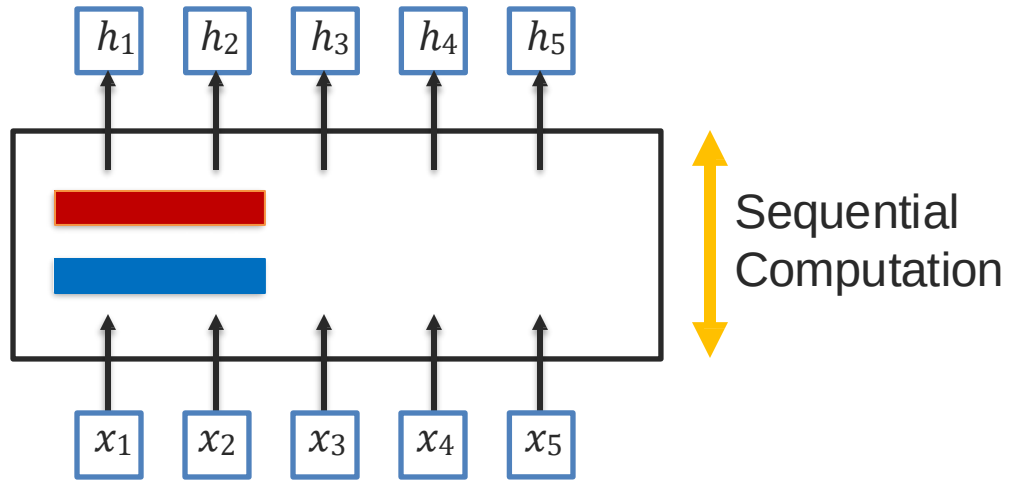
Ilya Sutskever at Neurips'24

Vision Transformer



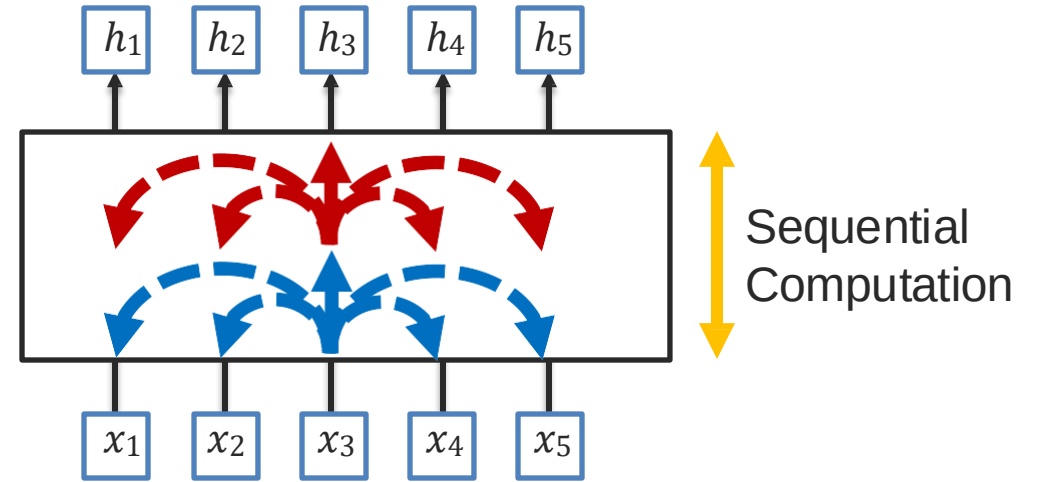
CNNs vs Transformers

Convolutions



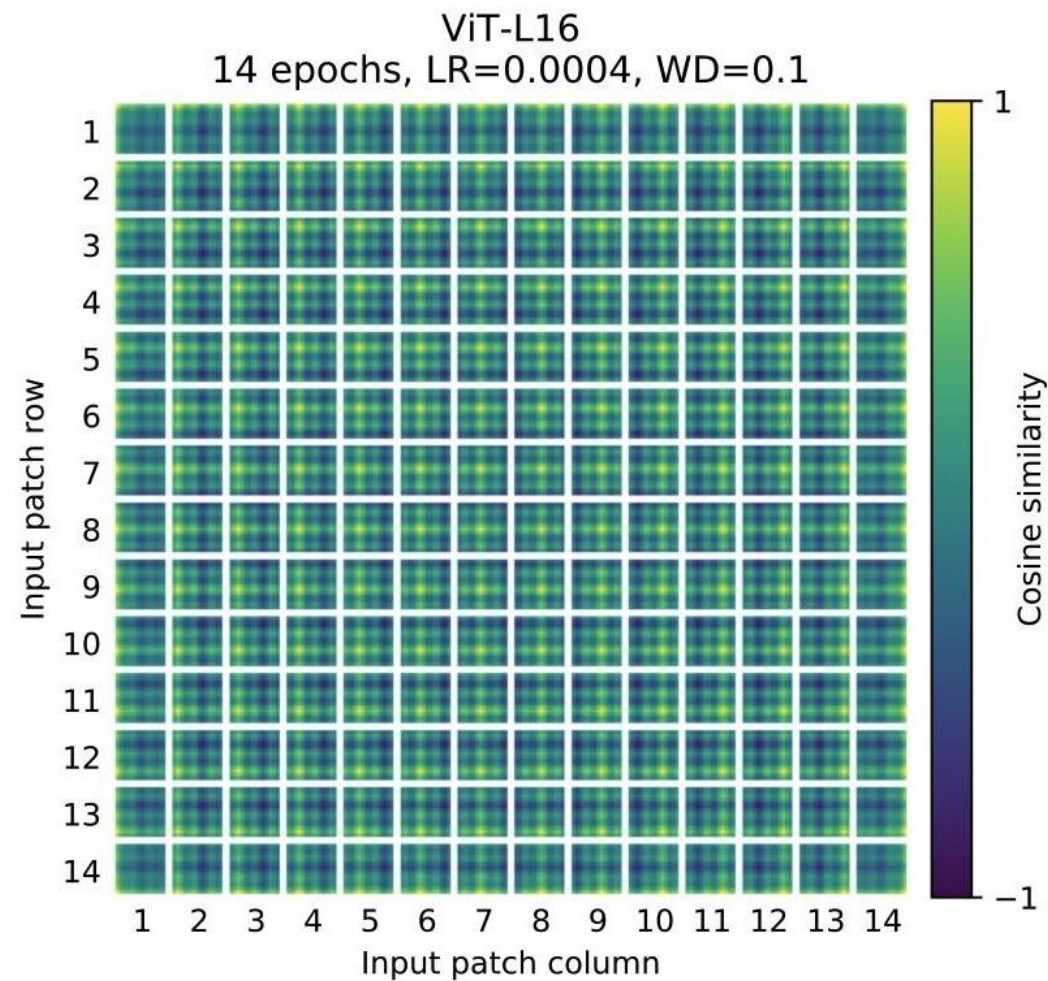
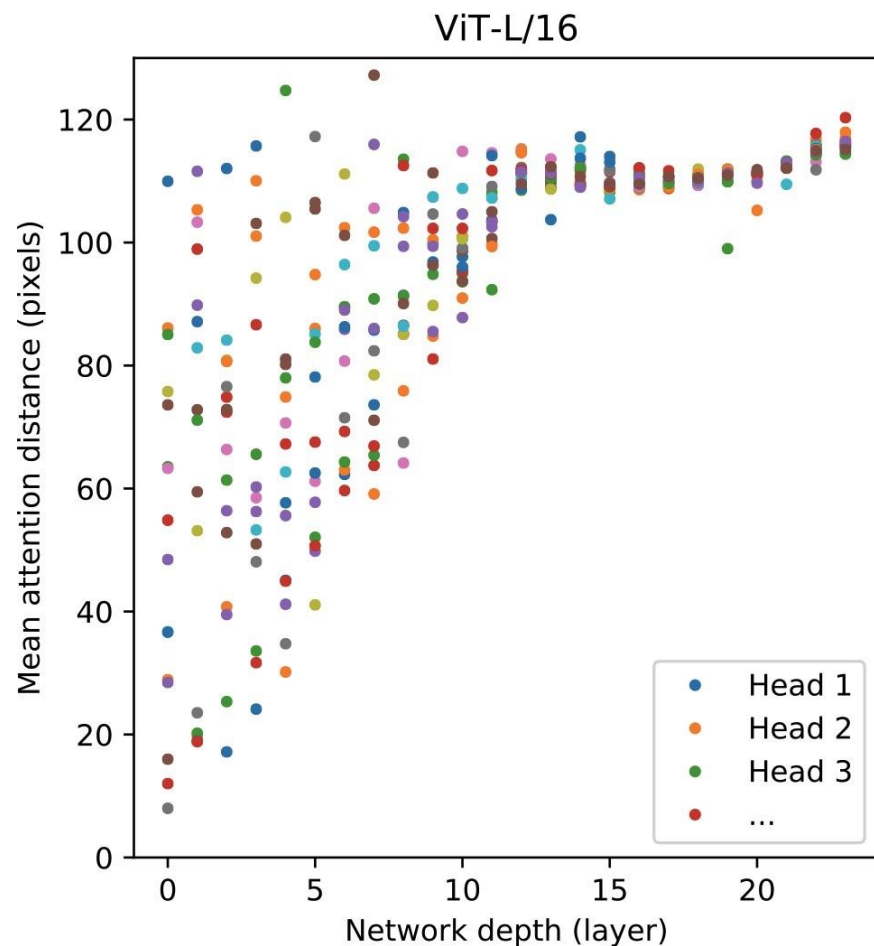
Modeling long-range dependencies
requires many layers.

Self-attention



No inductive bias toward
locality

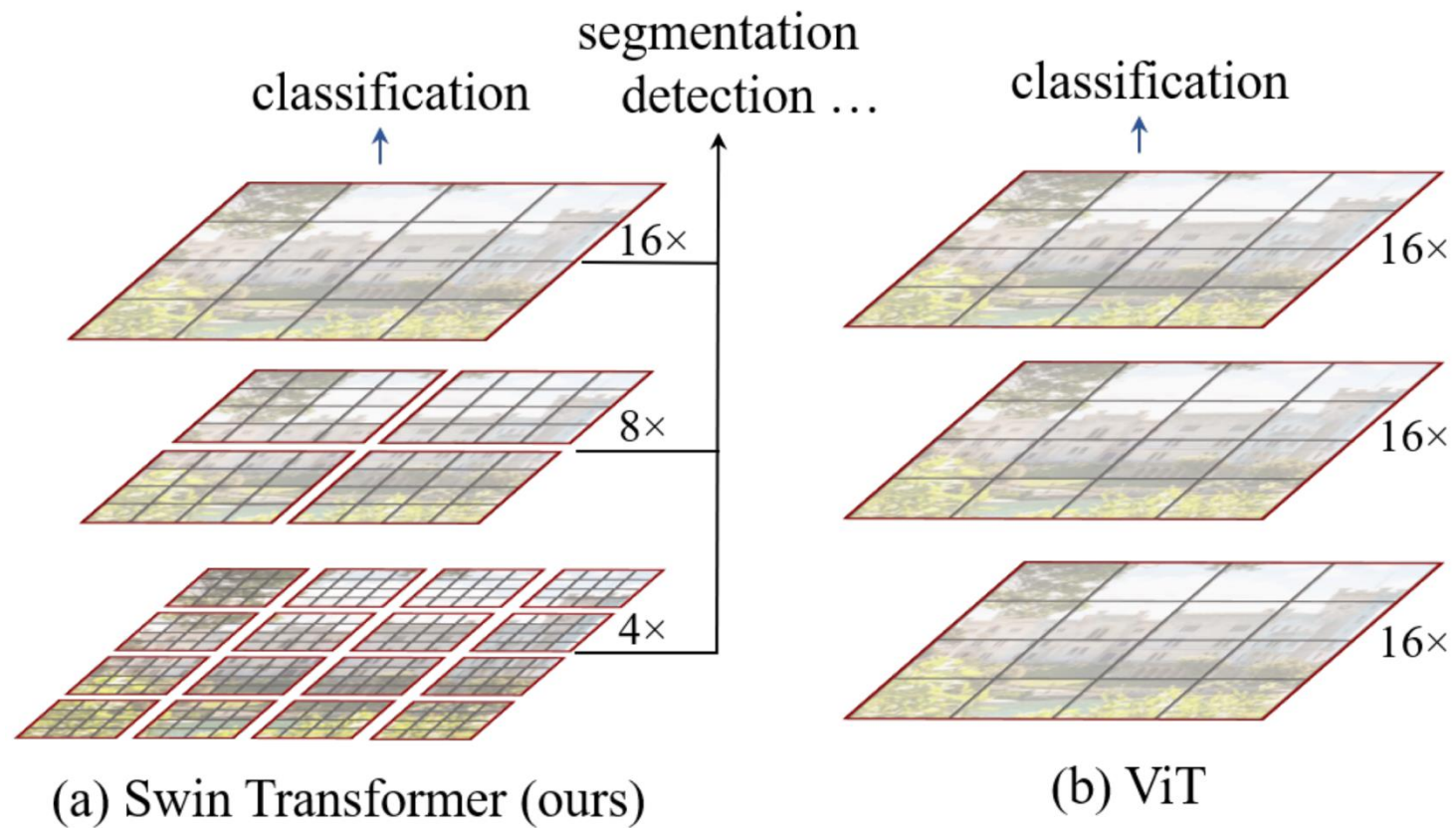
ViT do learn locality bias



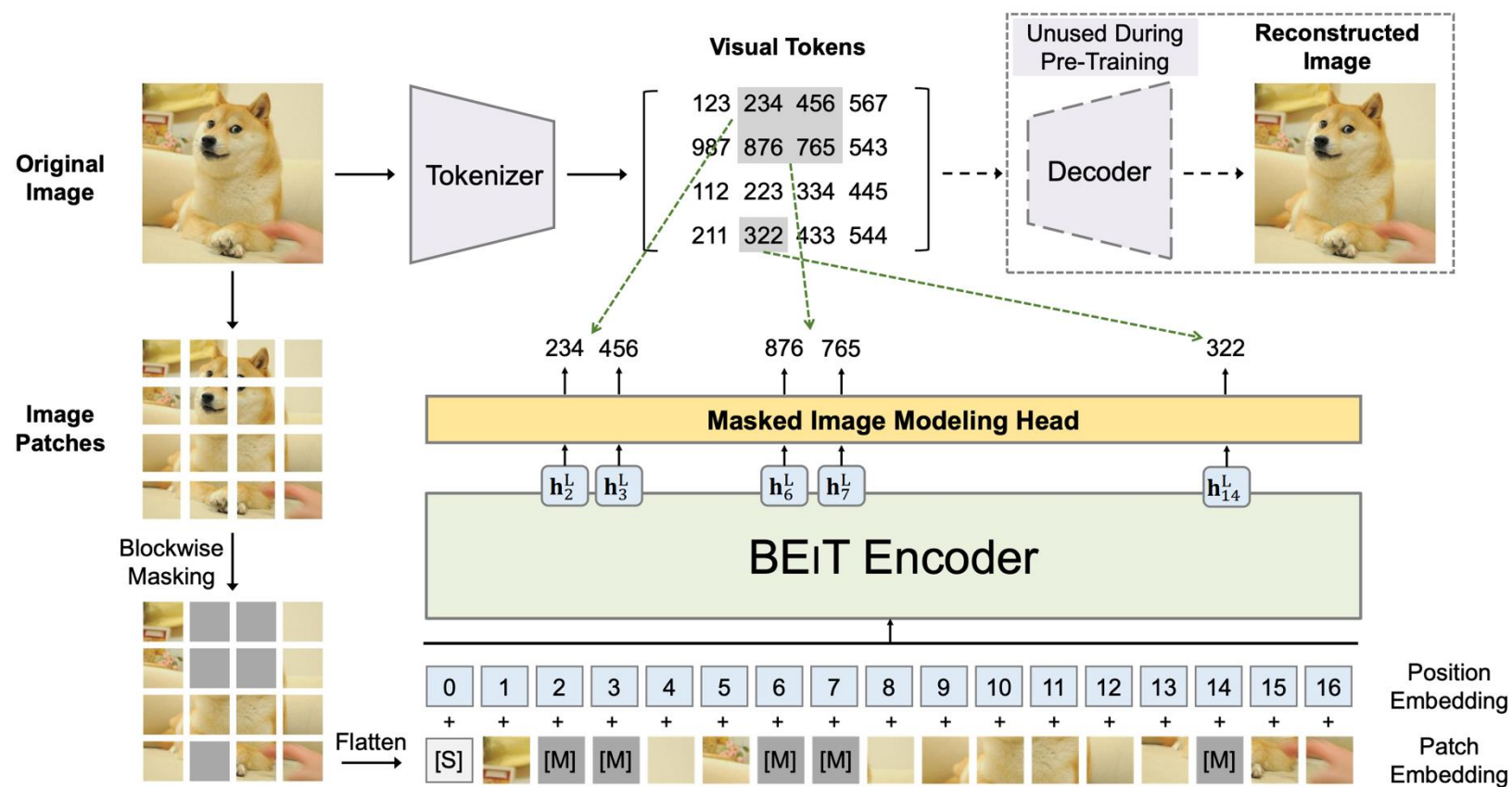
Which one is better? Approximately the same.

Model	Epochs	ImageNet	ImageNet ReaL	CIFAR-10	CIFAR-100	Pets	Flowers	exaFLOPs
ViT-B/32	7	80.73	86.27	98.61	90.49	93.40	99.27	164
ViT-B/16	7	84.15	88.85	99.00	91.87	95.80	99.56	743
ViT-L/32	7	84.37	88.28	99.19	92.52	95.83	99.45	574
ViT-L/16	7	86.30	89.43	99.38	93.46	96.81	99.66	2586
ViT-L/16	14	87.12	89.99	99.38	94.04	97.11	99.56	5172
ViT-H/14	14	88.08	90.36	99.50	94.71	97.11	99.71	12826
ResNet50x1	7	77.54	84.56	97.67	86.07	91.11	94.26	150
ResNet50x2	7	82.12	87.94	98.29	89.20	93.43	97.02	592
ResNet101x1	7	80.67	87.07	98.48	89.17	94.08	95.95	285
ResNet152x1	7	81.88	87.96	98.82	90.22	94.17	96.94	427
ResNet152x2	7	84.97	89.69	99.06	92.05	95.37	98.62	1681
ResNet152x2	14	85.56	89.89	99.24	91.92	95.75	98.75	3362
ResNet200x3	14	87.22	90.15	99.34	93.53	96.32	99.04	10212
R50x1+ViT-B/32	7	84.90	89.15	99.01	92.24	95.75	99.46	315
R50x1+ViT-B/16	7	85.58	89.65	99.14	92.63	96.65	99.40	855
R50x1+ViT-L/32	7	85.68	89.04	99.24	92.93	96.97	99.43	725
R50x1+ViT-L/16	7	86.60	89.72	99.18	93.64	97.03	99.40	2704
R50x1+ViT-L/16	14	87.12	89.76	99.31	93.89	97.36	99.11	5165

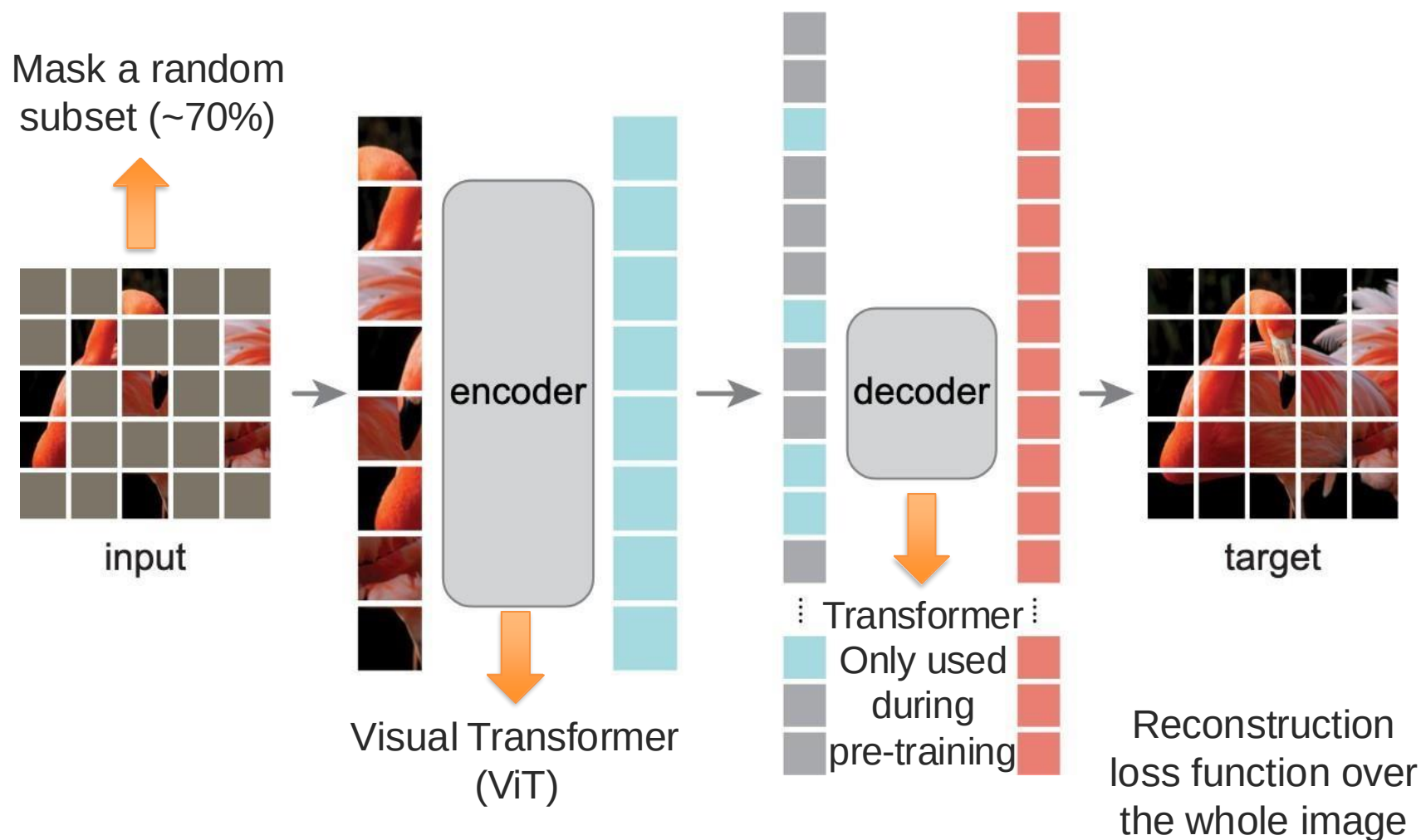
Hierarchical Vision Transformer



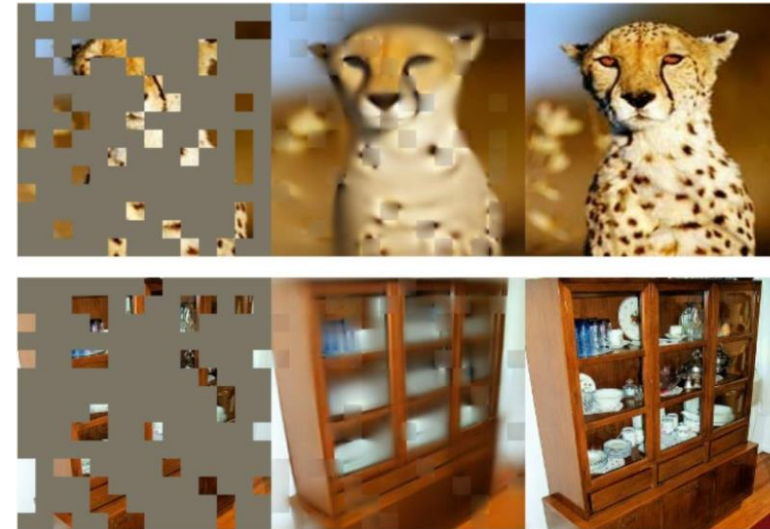
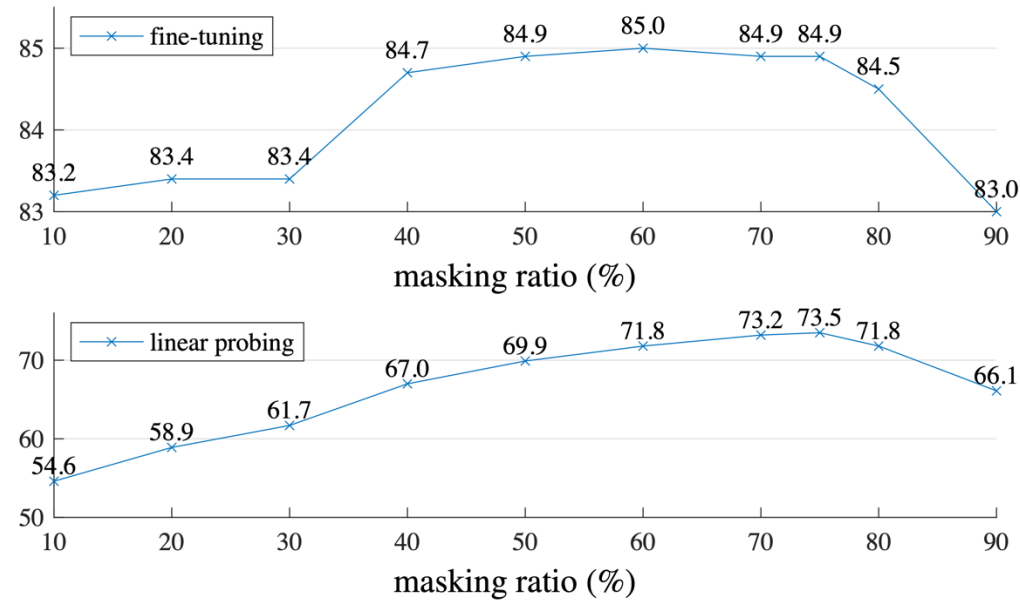
Visual Tokens



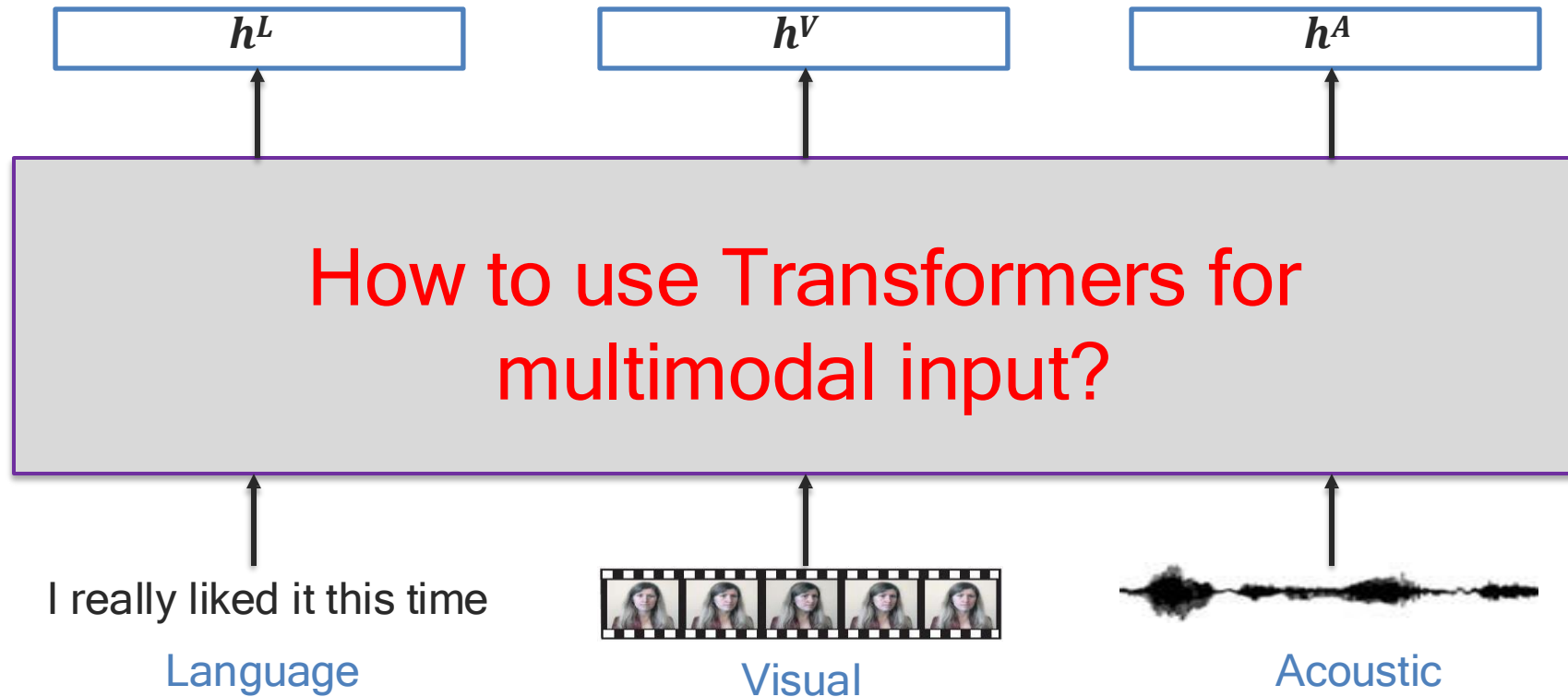
Masked Auto-Encoder (MAE)



Masked Auto-Encoder (MAE)



Multimodal Transformers

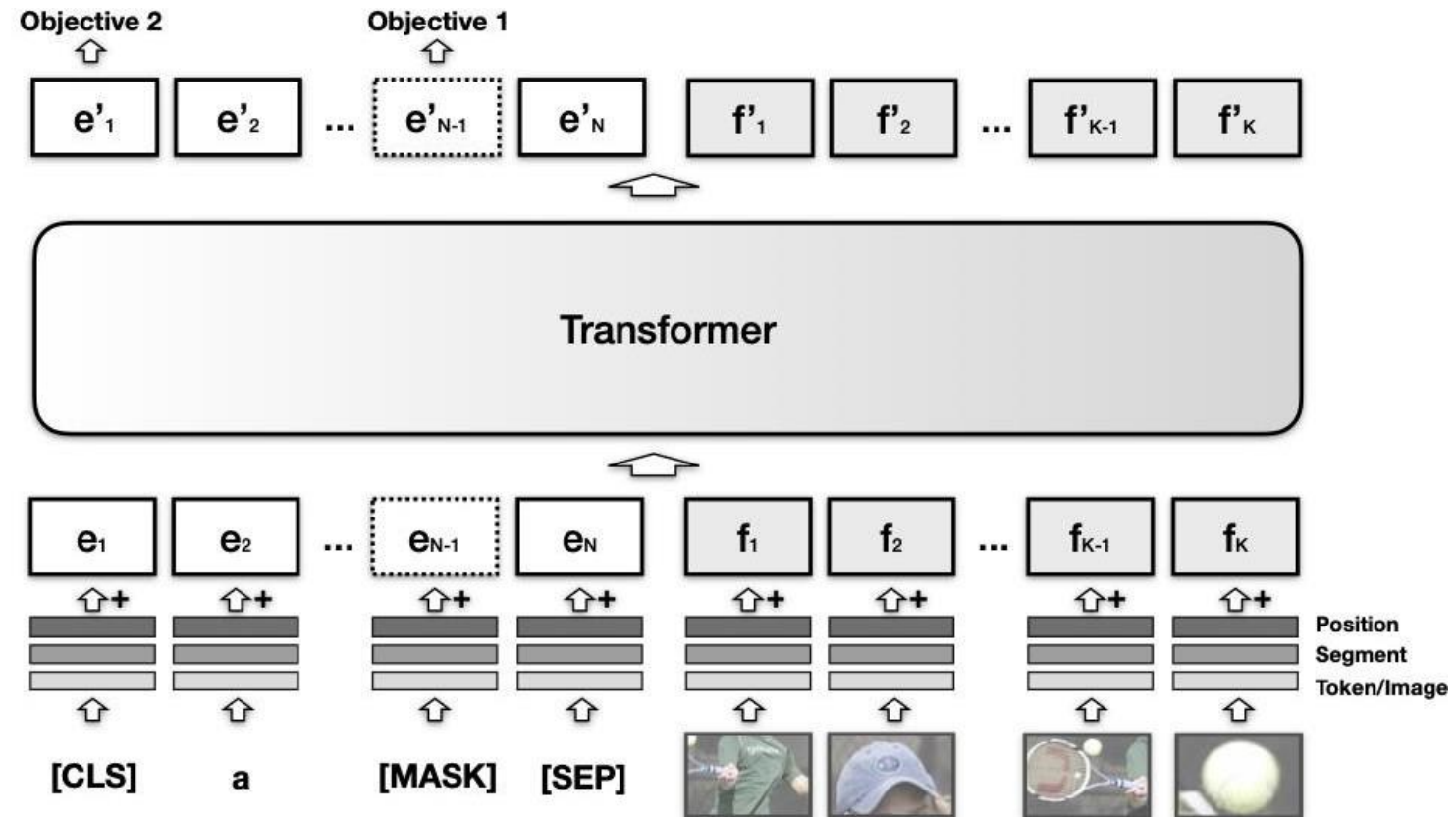


Option 1: Concatenate modalities and learn BERT transformer

VisualBERT



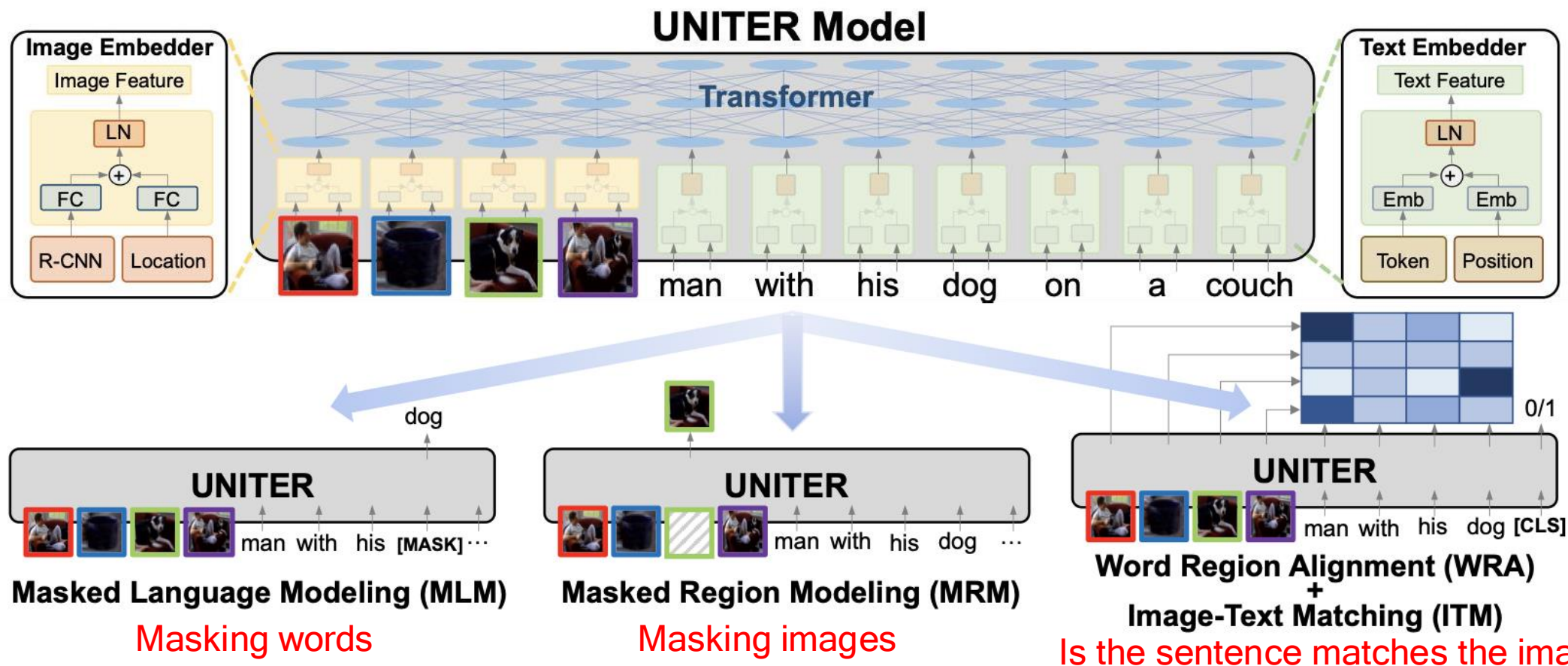
A person hits a ball with a tennis racket



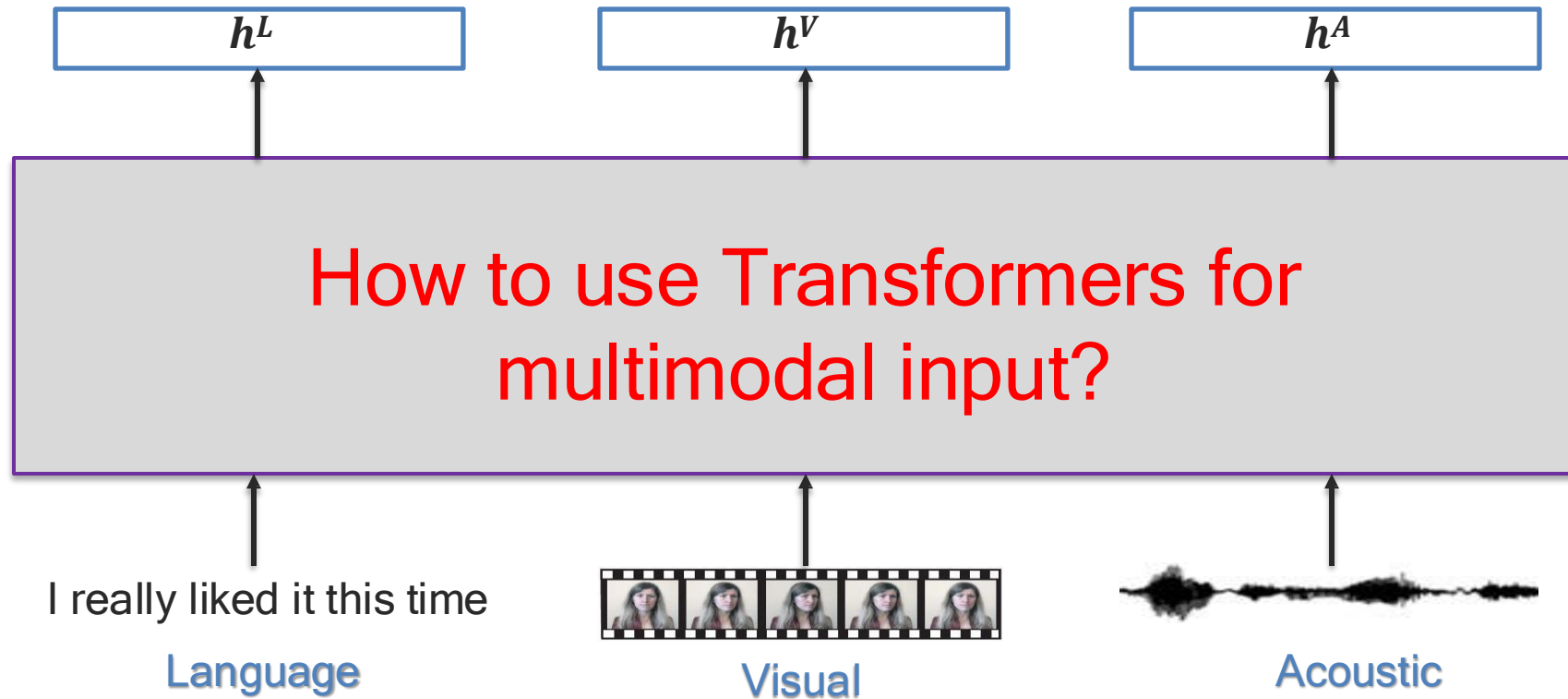
Li, Liunian Harold, et al. "Visualbert: A simple and performant baseline for vision and language. (2019)

UNITER

Similar Transformer architecture to BERT and VisualBERT... but with slightly different optimization

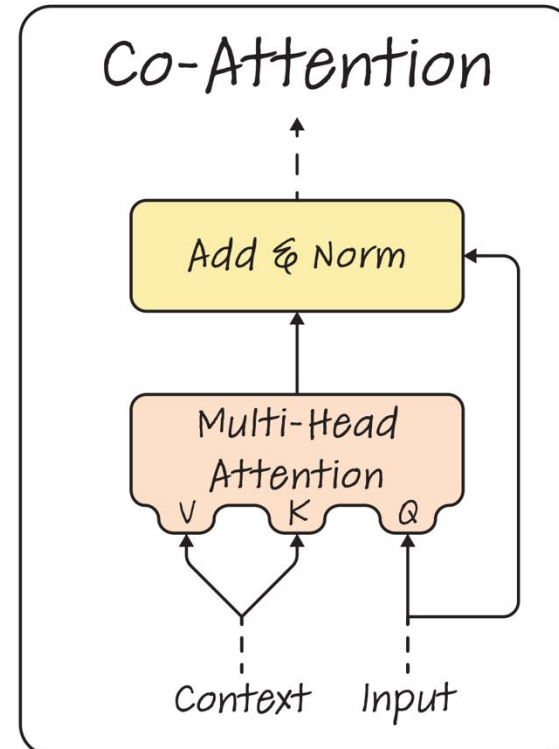
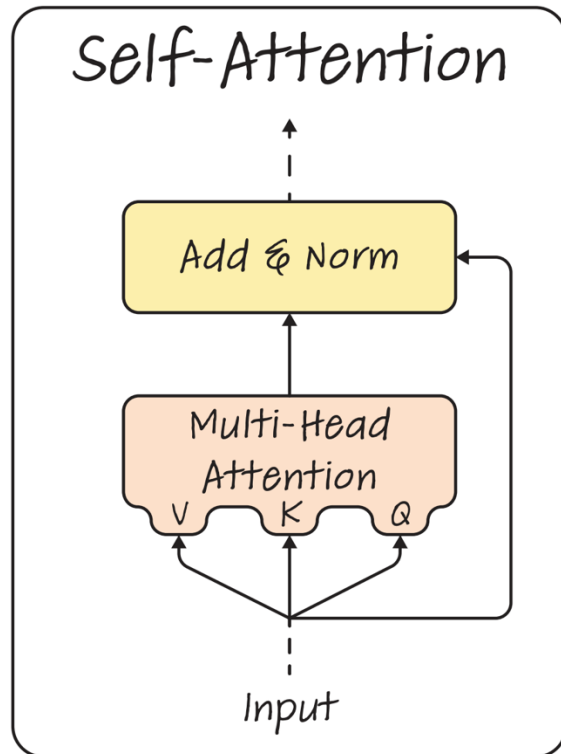


Multimodal Embeddings



Option 2: Look at pairwise interactions between modalities

Pairwise Cross-Attention



Cross-Attention

- Given two modalities $a \in \mathbb{R}^{n_A \times d}$, $b \in \mathbb{R}^{n_B \times d}$
- The attention score is based on a **dot-product**
 - $Q_A = a \cdot W_{A,Q} \in \mathbb{R}^{n_A \times d}$
 - $K_B = b \cdot W_{B,K} \in \mathbb{R}^{n_B \times d}$
 - Where: $W_{A,Q} \in \mathbb{R}^{d \times d}$, $W_{B,K} \in \mathbb{R}^{d \times d}$

Correlation Matrix: $Q_A \times K_B^T \in \mathbb{R}^{n_A \times n_B}$

Cross-attention correlation matrix

$$Q_A \quad Q_B^T$$
$$\begin{bmatrix} \square & \square & \square \\ \square & \square & \square \end{bmatrix} \times \begin{bmatrix} \square & \square \\ \square & \square \\ \square & \square \end{bmatrix} \in \mathbb{R}^{n_A \times n_B}$$

Correlation Matrix:

			
What	1.2	2.3	1.7
animal	-0.4	6.5	4.2
is	1.3	1.2	0.6

Co-Attention Transformer

1)

Co-correlation matrix
 $\mathbb{R}^{n_A \times n_B}$

$$\text{softmax} \left(\frac{Q_A \times K_B^T}{\sqrt{d_k}} \right) V_B$$

$\mathbb{R}^{n_B \times d}$

$Z_{B \rightarrow A}$

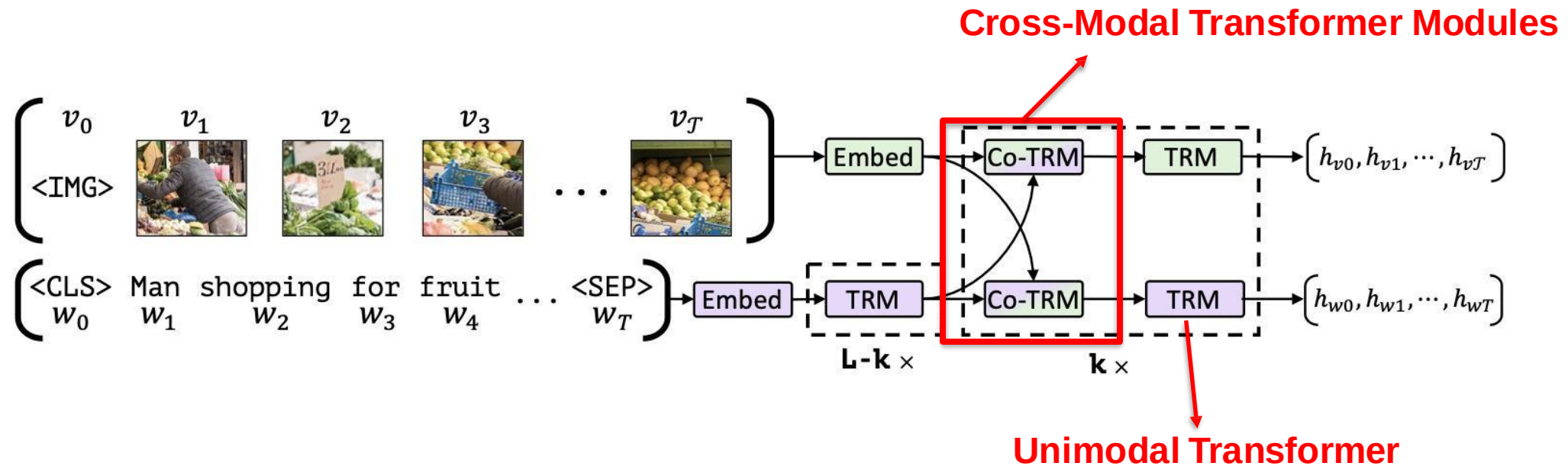
$\mathbb{R}^{n_A \times d}$

2)

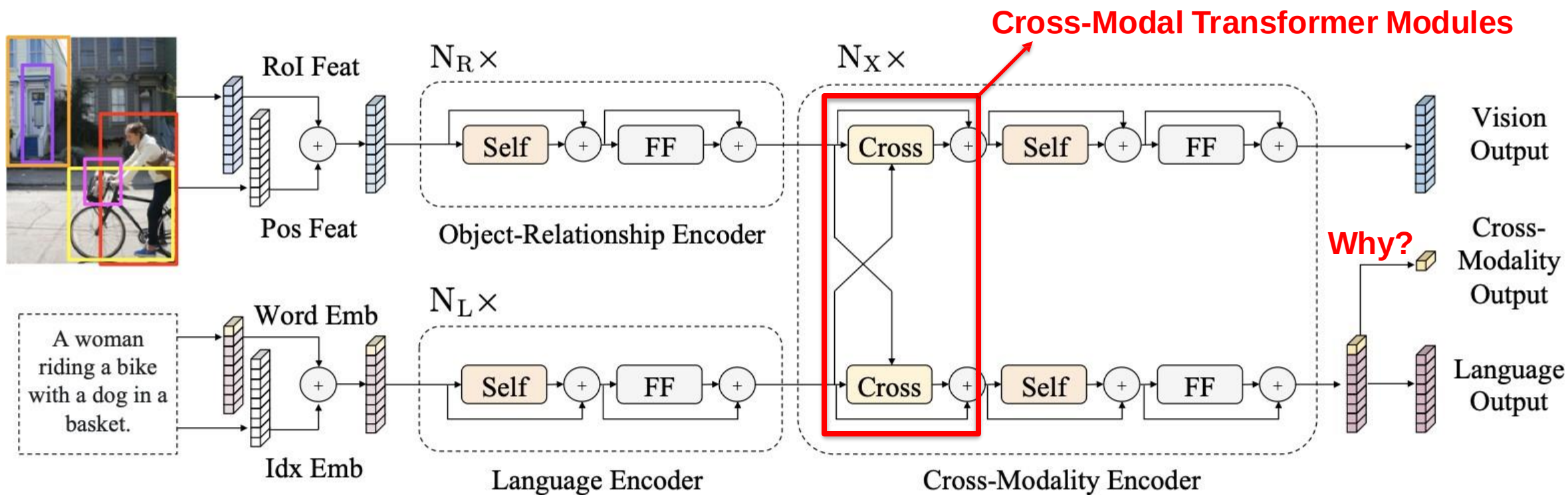
Residual connection

$Q_A + Z_{B \rightarrow A}$

ViLBERT

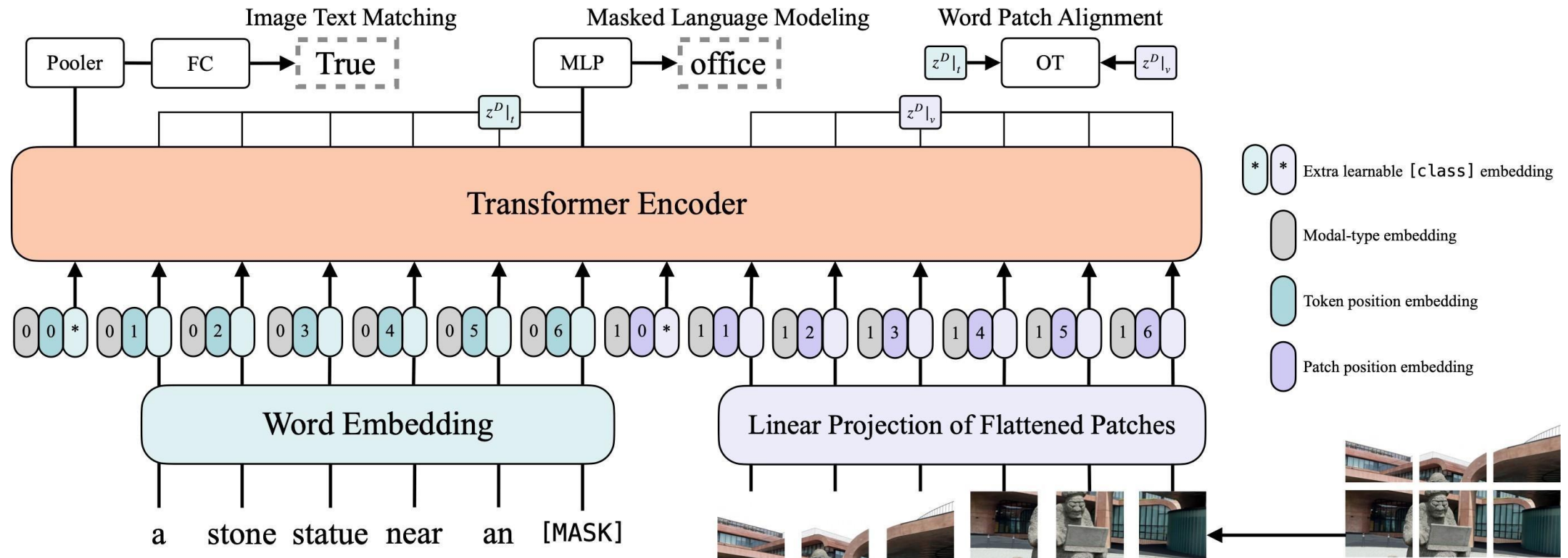


LXMERT



Tan, Hao, and Mohit Bansal. "Lxmert: Learning cross-modality encoder representations from transformers." *arXiv* (August 20, 2019).

Visual-and-Language Transformer (ViLT) ($\approx$ BERT + ViT)



Alignment between modalities



a display of **flowers** growing out and over the retaining **wall** in front of **cottages** on a **cloudy** day.



flowers



wall



cottages



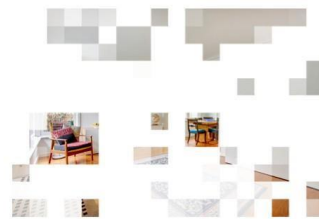
cloudy



a room with a **rug**, a **chair**, a **painting**, and a **plant**.



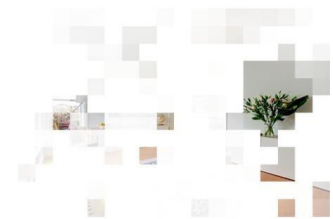
rug



chair



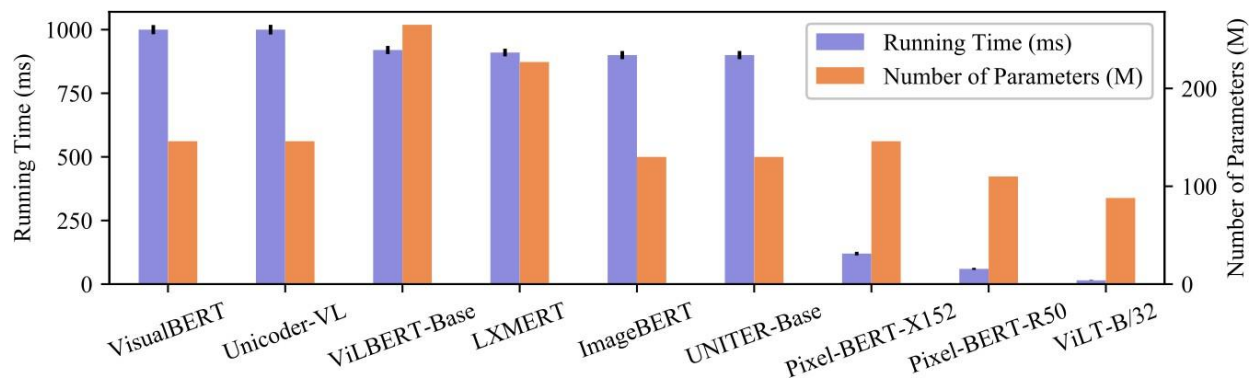
painting



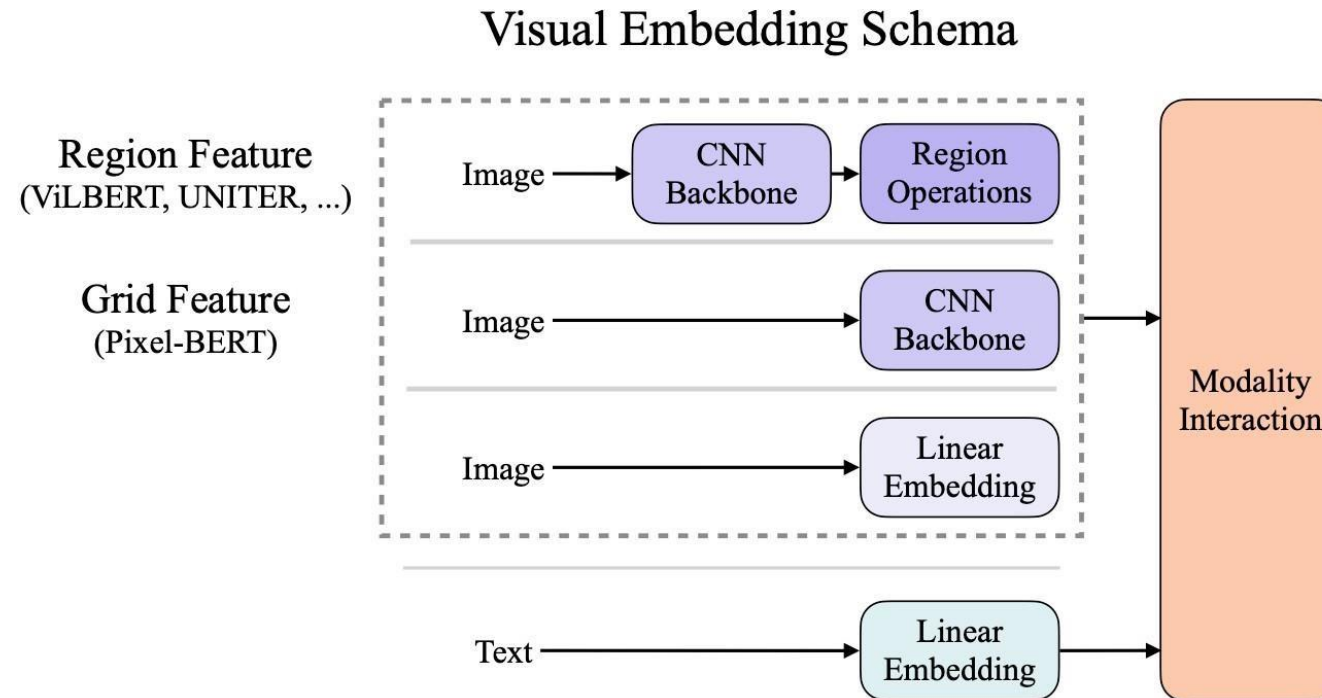
plant

ViLT: Faster Inference?

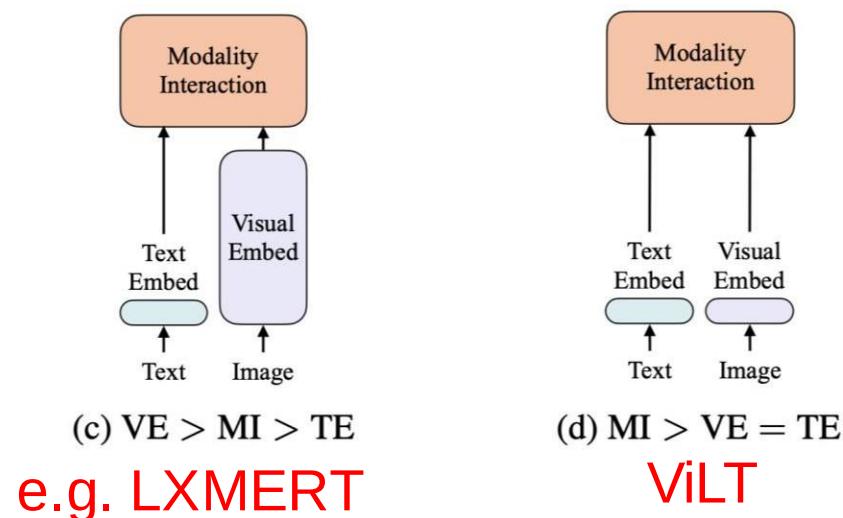
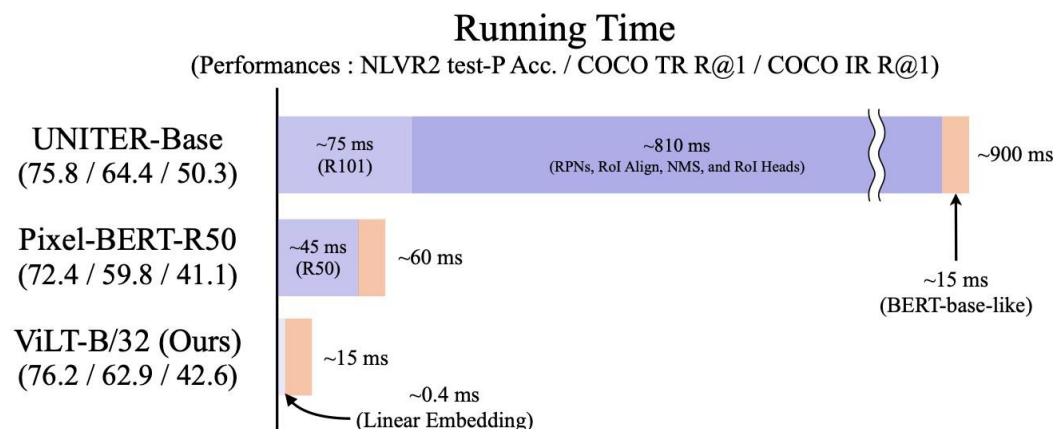
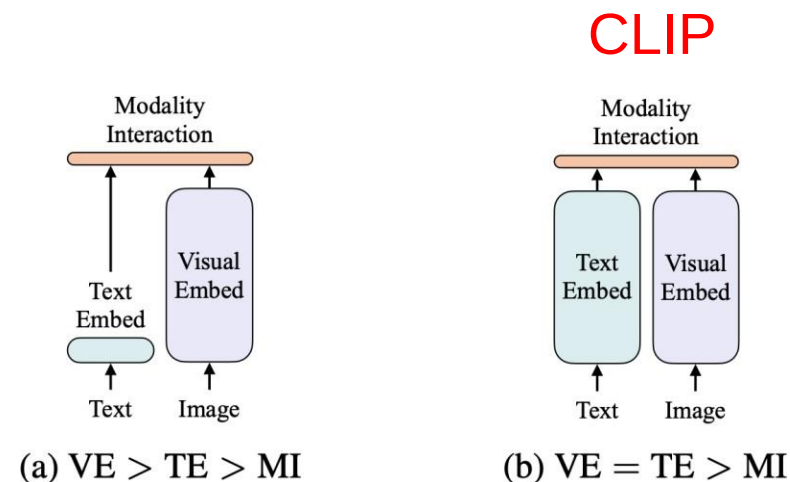
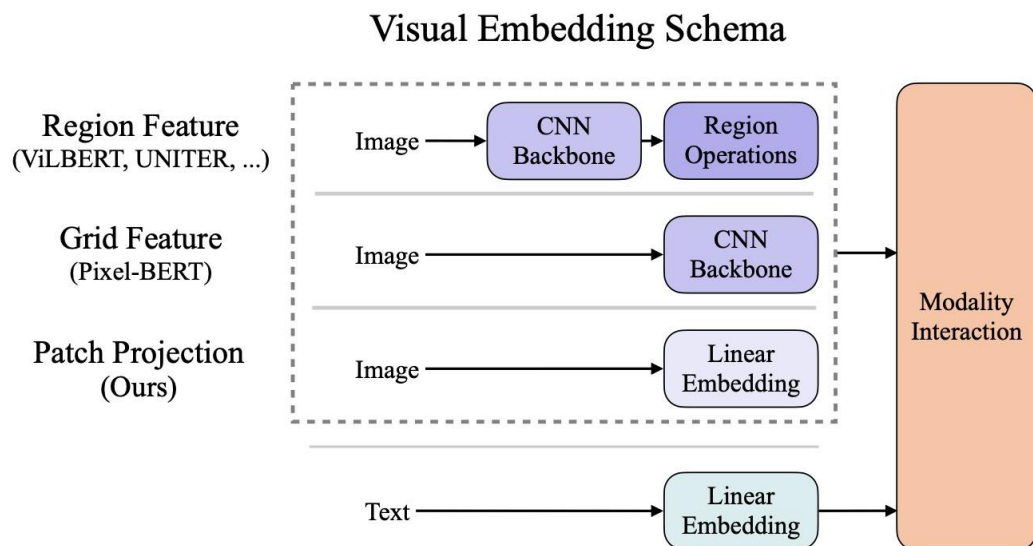
Visual Embed	Model	Time (ms)	VQAv2 test-dev	NLVR2 dev	NLVR2 test-P
Region	w/o VLP SOTA	~900	70.63	54.80	53.50
	ViLBERT	~920	70.55	-	-
	VisualBERT-Base	~1000	70.80	67.40	67.00
	LXMERT	~910	72.42	74.90	74.50
	UNITER-Base	~900	72.70	75.85	75.80
	OSCAR-Base [†]	~900	73.16	78.07	78.36
	VinVL-Base ^{†‡}	~1000	75.95	82.05	83.08
Grid	Pixel-BERT-X152	~120	74.45	76.50	77.20
	Pixel-BERT-R50	~60	71.35	71.70	72.40
Linear	ViLT-B/32	~15	70.34	74.56	74.66
	ViLT-B/32 [Ⓐ]	~15	70.94	75.24	76.21



Visual Transformers for Multimodal Learning



Visual Transformers for Multimodal Learning



Questions